

$\zeta(7)$ is irrational

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Abstract

We prove that $\zeta(7)$ is irrational. The proof constructs integer polynomials Q_n of degree $37n$ with

$$0 < Q_n(\zeta(7)) < \exp(-19000 n^2)$$

for all sufficiently large n . These polynomials are rationally normalized Hankel determinants attached to a positive moment functional. A positive integral representation gives nonvanishing; a logarithmic-energy estimate with respect to an explicit comparison measure gives the real bound; local p -adic determinant estimates over $\mathbb{Z}_p[X]$ control the coefficients after an explicit rational normalization. A perturbation estimate also gives $|\zeta(7) - a/b| > b^{-270}$ for every integer a and all sufficiently large positive integers b .

Contents

1	Introduction	2
1.1	The main theorem	3
1.2	Rational approximation	3
1.3	The construction and the proof	4
1.4	Organization	4
2	The determinant construction	5
2.1	Parameters and notation	5
2.2	The functional μ_X	5
2.3	The matrix and the normalized determinant	5
2.4	Degree	6
2.5	The positive integral representation	7
2.6	The main polynomial estimates	9
3	The local rational functional	9
3.1	The auxiliary functional τ_X and the pullback identity	9
3.2	Reflection and difference identities	10
3.3	Completion at a prime	12
3.4	The distribution formula	13
3.5	Small primes	15
4	Local estimates for the determinant	17
4.1	The inner range	17
4.2	The outer range	19
4.2.1	The rank decomposition	19
4.2.2	The rank lemma	20
4.2.3	The outer weights	21

5	Normalization and the prime sum	21
5.1	The normalization factor	22
5.2	The inner limiting function	22
5.3	The outer limiting function	23
5.4	The tail integral	24
5.5	The prime-sum bound	25
6	The real determinant	27
6.1	The external field	27
6.2	The comparison measure	27
6.3	Zero-mass logarithmic energy	27
6.4	Bound for a general configuration	28
6.5	Andréief's identity and scaling	29
6.6	The real bound	30
A	The comparison measure and its elementary bounds	30
A.1	The sixteen intervals	30
A.2	Elementary enclosures	31
A.3	The potential inequality	32
A.4	Energy and the norm constant	33
B	Completion of the irrationality proof	33
B.1	Proof of the main polynomial estimates	33
B.2	Proof of the main theorem	34
A	Exact arithmetic integrals	34
A.1	Evaluation of the inner integral	34
A.2	Evaluation of the outer integral	35
A.3	The final rational margins	35
B	Rational approximation and coefficient height	39
B.1	The relative matrix norm	39
B.2	Proof of the approximation bound	40
B.3	Coefficient height and primitive normalization	41

1 Introduction

Apéry proved in 1978 that $\zeta(3)$ is irrational [1]. For higher odd arguments, much of the subsequent progress has concerned families of values. Rivoal [11] and Ball–Rivoal [2] proved that infinitely many odd zeta values are irrational. Zudilin [13] showed that at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational. Later work has strengthened the lower bounds for the number of irrational values [7] and for the dimension of their rational span [6]. These theorems give information about the family as a whole, but do not determine the irrationality of any specified odd zeta value beyond $\zeta(3)$.

For $\zeta(5)$ itself, Brown and Zudilin [4] constructed an effective infinite sequence of rational approximations satisfying $0 < |\zeta(5) - p/q| < q^{-0.86}$. Their construction uses cellular integrals and the arithmetic of the associated linear forms. The exponent 0.86 does not by itself imply irrationality. To obtain irrationality from approximations of this kind, one needs a decay estimate

that remains sufficient after the denominators have been cleared. The same comparison between real decay and denominator growth governs the polynomial construction used below.

Recently, Fauzan [5] proved that $\zeta(5)$ is irrational by a Hankel-determinant construction with a positive moment representation and a logarithmic-energy comparison. The present paper extends that argument to $\zeta(7)$. The two changes in the construction are $D_N^6 \rightarrow D_N^8$ in the numerator of the entries of the Gram matrix, and the pole functional

$$j^4(X - H_j^{(5)}) - \frac{1}{4} + \frac{1}{2j} \longrightarrow j^6(X - H_j^{(7)}) + \frac{1}{2j} - \frac{1}{6}.$$

The rest of the argument follows the same architecture: a positive moment functional, a Hankel-type determinant, a positive integral representation, a logarithmic-energy comparison, local p -adic estimates, and a rational normalization that makes the determinant integral.

1.1 The main theorem

Theorem 1.1. *The number $\zeta(7)$ is irrational.*

The proof uses integer polynomials of growing degree. For each integer $M \in 40\mathbb{Z}_{>0}$ and each $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$, we construct a rational functional μ_X , an $h \times h$ Hankel-type matrix $G_K(X)$ with $h = 37K/40$, and a polynomial

$$Q_{K,M}(X) = m_{K,M} F_K(X),$$

where $F_K = S_K \Delta_K$ is a normalized determinant and $m_{K,M}$ is a rational normalization factor defined in Section 5.

Theorem 1.2. *For each integer $M \in 40\mathbb{Z}_{>0}$ and each $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$, the polynomial $Q_{K,M}$ belongs to $\mathbb{Z}[X]$, has degree h , and satisfies $Q_{K,M}(\zeta(7)) > 0$. For all sufficiently large positive integers n ,*

$$0 < Q_{40n,200}(\zeta(7)) < \exp(-19000 n^2), \tag{1}$$

$$0 < Q_{40n,100000}(\zeta(7)) < \exp(-19000 n^2). \tag{2}$$

The choice $M = 200$ suffices for Theorem 1.1. The larger cutoff provides the additional decay used in the approximation bound of Corollary 1.3. The proof of Theorem 1.2 is given in Section B.

1.2 Rational approximation

Corollary 1.3. *There is a positive integer b_0 such that*

$$\left| \zeta(7) - \frac{a}{b} \right| > b^{-270}$$

for every $a \in \mathbb{Z}$ and every $b \geq b_0$.

The proof is given in Appendix B.

1.3 The construction and the proof

The polynomial estimates of Theorem 1.2 imply Theorem 1.1 directly. Indeed, suppose $\zeta(7) = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$. Then $b^{37n} Q_{40n,200}(a/b)$ is a positive integer. On the other hand, (1) gives

$$0 < b^{37n} Q_{40n,200}(a/b) < \exp(37n \log b - 19000n^2),$$

and the right-hand side tends to zero as $n \rightarrow \infty$. This is impossible for an integer. Hence $\zeta(7) \notin \mathbb{Q}$.

We now describe the construction behind $Q_{K,M}$. The starting point is a $\mathbb{Q}$ -linear functional μ_X on rational functions with simple poles among the negative integer squares $\{-j^2 : j \geq 1\}$. Its values on monomials t^e are the Bernoulli numbers divided by the appropriate factorial, and its values on simple poles $1/(t + j^2)$ are affine in X , of the form $j^6(X - H_j^{(7)}) + 1/(2j) - 1/6$. At $X = \zeta(7)$, the functional becomes integration against a positive weight $w_7(y)$. This positive integral representation is the source of the nonvanishing of the determinant.

Apply μ_X to the rational functions $D_N(t)^8 t^{i+j}/D_K(t)$ with $D_m(t) = \prod_{j=1}^m (t + j^2)$, and form the $h \times h$ matrix

$$G_K(X) = \left[\mu_X \left(\frac{D_N(t)^8 t^{i+j}}{D_K(t)} \right) \right]_{0 \leq i, j < h}.$$

Each entry is affine in X , so $\Delta_K(X) = \det G_K(X)$ is a polynomial in X of degree h . The matrix $G_K(\zeta(7))$ is a positive definite Gram matrix, so $\Delta_K(\zeta(7)) > 0$.

Two distinct estimates complete the proof. First, an arithmetic estimate controls every coefficient of $\Delta_K(X)$. This is done by reducing to p -adic completions at each prime, extending μ_X to a completed functional τ_X^{ext} on the Tate algebra, and bounding the local determinant valuations in a suitable basis. The normalization factor S_K (a product of factorials) and the additional factor $m_{K,M}$ (a product of prime powers) clear all denominators and make the polynomial integral.

Second, a real estimate controls the size of $F_K(\zeta(7)) = S_K \Delta_K(\zeta(7))$. By Andréief's identity, $\Delta_K(\zeta(7))$ is a multiple integral over h variables, with a squared Vandermonde factor. After scaling, the integrand is the exponential of a discrete logarithmic energy associated with an external field $V(t)$. Comparing this energy to the equilibrium measure ρ of mass $\lambda = 37/40$, we obtain the required real bound on the K^2 scale.

Both estimates are balanced against the normalization factor $m_{K,M}$, and the final inequality

$$\limsup_{K \rightarrow \infty} K^{-2} \log Q_{K,M}(\zeta(7)) \leq A_M + \overline{U} < 0$$

follows from the explicit constants A_M (from the prime sum) and $\overline{U}$ (from the real bound) computed in Sections 5 and 6. The negative quadratic decay then yields Theorem 1.2.

For Corollary 1.3, smallness of $Q_{K,M}(\zeta(7))$ alone is not enough, since the polynomial could vanish at a nearby rational point. We bound the coefficient matrix of X relative to the positive definite matrix $G_K(\zeta(7))$. This preserves positivity under the required perturbation and permits the integer argument to be applied at the rational point. The larger fixed cutoff $M = 100000$ provides the additional decay for that argument.

1.4 Organization

Section 2 defines the matrix and proves its degree and positivity. Section 3 introduces the local rational functional τ_X , establishes the completion and distribution formulas, and proves the small-prime estimate. Section 4 establishes the inner and outer local estimates for the determinant. Section 5 combines the local bounds into a rational normalization and estimates the prime sum. Section 6 proves the real bound on the determinant via logarithmic energy. Section B combines the

two bounds and proves Theorems 1.1 and 1.2. Appendix A contains the comparison measure and its elementary bounds. Appendix A contains the exact arithmetic integrals. Appendix B proves Corollary 1.3.

2 The determinant construction

2.1 Parameters and notation

Throughout the paper,

$$K = 40n, \quad N = 3n, \quad h = 37n, \quad \alpha = \frac{3}{40}, \quad \lambda = \frac{37}{40}, \quad H = 1 + 3\alpha = \frac{49}{40}. \quad (3)$$

For $m \geq 0$, set

$$D_m(t) = \prod_{j=1}^m (t + j^2), \quad D_0(t) = 1. \quad (4)$$

Let $H_j^{(7)} = \sum_{v=1}^j v^{-7}$ for $j \geq 1$, and $H_0^{(7)} = 0$. Let $(B_k)_{k \geq 0}$ denote the Bernoulli numbers, defined by

$$B_0 = 1, \quad \sum_{k=0}^n \binom{n+1}{k} B_k = 0 \quad (n \geq 1),$$

so that $B_1 = -1/2$, $B_2 = 1/6$, $B_3 = 0$, $B_4 = -1/30$, $B_5 = 0$, $B_6 = 1/42$, and in general $B_{2k+1} = 0$ for $k \geq 1$.

Throughout, $v_p : \mathbb{Q}_p^\times \rightarrow \mathbb{Z}$ is the normalized p -adic valuation. For a polynomial $A \in \mathbb{Q}_p[X]$, we write $v_p^G(A)$ for the minimum of the p -adic valuations of its coefficients, with $v_p^G(0) = +\infty$.

2.2 The functional μ_X

Definition 2.1. Let μ_X be the $\mathbb{Q}$ -linear functional on the $\mathbb{Q}$ -vector space of rational functions with only simple poles among $\{-j^2 : j \geq 1\}$ defined by

$$\mu_X(t^e) = \frac{(-1)^e B_{2e+2} (2e+3)(2e+4)(2e+5)(2e+6)(2e+7)}{720} \quad (e \geq 0), \quad (5)$$

$$\mu_X\left(\frac{1}{t+j^2}\right) = j^6(X - H_j^{(7)}) + \frac{1}{2j} - \frac{1}{6} \quad (j \geq 1). \quad (6)$$

Extended to all such rational functions by polynomial division and simple partial fraction decomposition.

The values $\mu_X(t^e)$ are independent of X ; the values on the simple poles are affine in X . The choice of coefficients in (5)–(6) is dictated by Proposition 2.4: it ensures that at $X = \zeta(7)$ the functional becomes integration against a positive weight $w_7(y)$.

2.3 The matrix and the normalized determinant

Definition 2.2. Define

$$G_K(X) = \left[\mu_X\left(\frac{D_N(t)^8 t^{i+j}}{D_K(t)}\right) \right]_{0 \leq i, j < h}, \quad \Delta_K(X) = \det G_K(X). \quad (7)$$

Define

$$S_K = \frac{(K!)^{2h} \cdot 4^{h-1}}{(N!)^{16h} \cdot \prod_{i=1}^{h-1} ((2i)!)^2}, \quad F_K = S_K \Delta_K. \quad (8)$$

For a fixed integer $M \geq 40$ and $K \geq 200M^2$, let $m_{K,M} > 0$ be the positive rational number defined by (61) in Section 5. Define

$$Q_{K,M}(X) = m_{K,M} F_K(X) = m_{K,M} S_K \Delta_K(X). \quad (9)$$

2.4 Degree

Lemma 2.3. *For every $K \in 40\mathbb{Z}_{>0}$, the determinant $\Delta_K(X)$ is a polynomial of degree exactly h , with leading coefficient*

$$[X^h] \Delta_K(X) = (-1)^{h(h-1)/2} \prod_{j=N+1}^K j^6 D_N(-j^2)^7 \neq 0. \quad (10)$$

Consequently, for each fixed $M \geq 40$ and $K \geq 200M^2$, the polynomial $Q_{K,M}$ also has degree h .

Proof. Set $D_{\text{tail}}(t) = D_K(t)/D_N(t) = \prod_{j=N+1}^K (t + j^2)$. The rational function

$$R_{ij}(t) = \frac{D_N(t)^8 t^{i+j}}{D_K(t)} = \frac{D_N(t)^7 t^{i+j}}{D_{\text{tail}}(t)}$$

has only simple poles, at $t = -j^2$ for $N < j \leq K$, and no other poles. Polynomial division and simple partial fraction decomposition give

$$R_{ij}(t) = P_{ij}(t) + \sum_{j'=N+1}^K \frac{c_{ij,j'}}{t + (j')^2}, \quad (11)$$

with $P_{ij} \in \mathbb{Q}[t]$ and $c_{ij,j'} \in \mathbb{Q}$. The polynomial part contributes $\mu_X(P_{ij})$, which is independent of X . The simple pole at $t = -(j')^2$ contributes

$$c_{ij,j'} \mu_X \left(\frac{1}{t + (j')^2} \right) = c_{ij,j'} \left[(j')^6 (X - H_{j'}^{(7)}) + \frac{1}{2j'} - \frac{1}{6} \right],$$

which is affine in X with X -coefficient $c_{ij,j'}(j')^6$. Hence each entry of $G_K(X)$ is affine in X , and

$$G_K(X) = A + XB, \quad \Delta_K(X) = \det(A + XB) \in \mathbb{Q}[X],$$

with $A, B \in M_h(\mathbb{Q})$.

The X -coefficient matrix is

$$B_{ij} = \sum_{j'=N+1}^K (j')^6 \operatorname{Res}_{t=-(j')^2} R_{ij}(t),$$

and the residue is

$$\operatorname{Res}_{t=-(j')^2} R_{ij}(t) = \frac{D_N(-(j')^2)^7 (-(j')^2)^{i+j}}{D'_{\text{tail}}(-(j')^2)}.$$

Setting $v_a = -a^2$ and

$$\gamma_a = \frac{a^6 D_N(-a^2)^7}{D'_{\text{tail}}(-a^2)} \quad (N < a \leq K),$$

we obtain

$$B_{ij} = \sum_{a=N+1}^K \gamma_a v_a^{i+j}, \quad B = V \operatorname{diag}(\gamma_a) V^T, \quad (12)$$

where $V = (v_a^i)_{0 \leq i < h, N < a \leq K}$ is the $h \times h$ Vandermonde matrix.

Since $K - N = h$ and the v_a are distinct, V is invertible. Its determinant is

$$\det V = \prod_{N < a < b \leq K} (v_b - v_a) = \prod_{N < a < b \leq K} (a^2 - b^2) \neq 0. \quad (13)$$

Furthermore,

$$D'_{\text{tail}}(-a^2) = \prod_{\substack{N < b \leq K \\ b \neq a}} (b^2 - a^2),$$

and each unordered pair $\{a, b\}$ contributes $(b^2 - a^2)$ from the term $D'_{\text{tail}}(-a^2)$ and $(a^2 - b^2) = -(b^2 - a^2)$ from the term $D'_{\text{tail}}(-b^2)$. Hence

$$\prod_{a=N+1}^K D'_{\text{tail}}(-a^2) = (-1)^{h(h-1)/2} (\det V)^2. \quad (14)$$

Therefore

$$\det B = (\det V)^2 \prod_{a=N+1}^K \gamma_a = (-1)^{h(h-1)/2} \prod_{a=N+1}^K a^6 D_N(-a^2)^7. \quad (15)$$

Each factor $D_N(-a^2) = \prod_{m=1}^N (m^2 - a^2)$ is nonzero because $a > N$ and $m \leq N$ imply $m^2 < a^2$. Hence $\det B \neq 0$, and $\deg \Delta_K = h$. Multiplication by $m_{K,M} S_K \neq 0$ preserves the degree. $\square$

2.5 The positive integral representation

Proposition 2.4. *Let*

$$w_7(y) = \frac{(2\pi)^6}{360} y^7 \sum_{\ell \geq 1} \ell^6 e^{-2\pi \ell y}, \quad y > 0. \quad (16)$$

Then $w_7(y) > 0$ for all $y > 0$, and for every rational function R in the domain of μ_X ,

$$\mu_{\zeta(7)}(R) = \int_0^\infty R(y^2) w_7(y) dy. \quad (17)$$

Consequently $G_K(\zeta(7))$ is positive definite, and $Q_{K,M}(\zeta(7)) > 0$ whenever $m_{K,M} > 0$.

Proof. The positivity $w_7(y) > 0$ for $y > 0$ is immediate from the definition.

Even moments. Fix $e \geq 0$. Integrating term by term,

$$\int_0^\infty y^{2e} w_7(y) dy = \frac{(2\pi)^6}{360} \sum_{\ell \geq 1} \ell^6 \int_0^\infty y^{2e+7} e^{-2\pi \ell y} dy.$$

The inner integral equals $(2e+7)!/(2\pi\ell)^{2e+8}$, so

$$\int_0^\infty y^{2e} w_7(y) dy = \frac{(2e+7)!}{360 \cdot (2\pi)^{2e+2}} \zeta(2e+2).$$

By Euler's formula $\zeta(2e+2) = (-1)^e B_{2e+2} (2\pi)^{2e+2} / (2(2e+2)!)$, we obtain

$$\int_0^\infty y^{2e} w_7(y) dy = \frac{(-1)^e B_{2e+2} (2e+3)(2e+4)(2e+5)(2e+6)(2e+7)}{720}, \quad (18)$$

matching $\mu_X(t^e)$ from (5).

Pole identity. Put $f(y) = (e^{2\pi y} - 1)^{-1}$, so that

$$w_7(y) = \frac{(2\pi)^6}{360} y^7 \sum_{\ell \geq 1} \ell^6 e^{-2\pi \ell y} = \frac{y^7}{360} f^{(6)}(y).$$

Six integrations by parts in the integral $\int_0^\infty w_7(y)/(y^2 + a^2) dy$, using vanishing boundary terms at 0 and $+\infty$ (at zero each boundary product is $O(y)$, at infinity each decays exponentially), give

$$\int_0^\infty \frac{w_7(y)}{y^2 + a^2} dy = \frac{1}{360} \int_0^\infty f(y) \frac{d^6}{dy^6} \left(\frac{y^7}{y^2 + a^2} \right) dy.$$

A direct computation gives

$$\frac{1}{360} \cdot \frac{d^6}{dy^6} \left(\frac{y^7}{y^2 + a^2} \right) = 2a^6 \frac{y(7a^6 - 35a^4 y^2 + 21a^2 y^4 - y^6)}{(y^2 + a^2)^7}.$$

Indeed, expanding $y^7/(y^2 + a^2) = y^5 - a^2 y^3 + a^4 y - a^6 y/(y^2 + a^2)$, the polynomial part has degree $5 < 6$ and therefore vanishes after six differentiations. For the last term, the identity

$$\frac{d^6}{dy^6} \left(\frac{y}{y^2 + a^2} \right) = -\frac{720 y(7a^6 - 35a^4 y^2 + 21a^2 y^4 - y^6)}{(y^2 + a^2)^7}$$

is verified by direct differentiation, giving the claimed expression.

Hermite's integral formula for the Hurwitz zeta function at $s = 7$ (see [10, DLMF 25.11.29]) reads

$$\zeta(7, a) = \frac{1}{2a^7} + \frac{1}{6a^6} + 2 \int_0^\infty \frac{y(7a^6 - 35a^4 y^2 + 21a^2 y^4 - y^6)}{(y^2 + a^2)^7} \frac{dy}{e^{2\pi y} - 1},$$

valid for $\Re a > 0$. Substituting gives

$$\int_0^\infty \frac{w_7(y)}{y^2 + a^2} dy = a^6 \zeta(7, a) - \frac{1}{2a} - \frac{1}{6}. \quad (19)$$

At $a = j \in \mathbb{Z}_{>0}$, using $\zeta(7, j) = \zeta(7) - H_j^{(7)} + j^{-7}$, we get

$$j^6 \zeta(7, j) - \frac{1}{2j} - \frac{1}{6} = j^6 (\zeta(7) - H_j^{(7)}) + \frac{1}{j} - \frac{1}{2j} - \frac{1}{6} = j^6 (\zeta(7) - H_j^{(7)}) + \frac{1}{2j} - \frac{1}{6},$$

matching (6) at $X = \zeta(7)$. By linearity, (17) holds for every rational function in the domain of μ_X .

Positivity. For a nonzero real polynomial $q(t) = \sum_{i=0}^{h-1} c_i t^i$ of degree less than h , the quadratic form of $G_K(\zeta(7))$ at the coefficient vector $c = (c_0, \dots, c_{h-1})$ is

$$c^T G_K(\zeta(7)) c = \mu_{\zeta(7)} \left(\frac{D_N(t)^8 q(t)^2}{D_K(t)} \right) = \int_0^\infty \frac{D_N(y^2)^8 q(y^2)^2}{D_K(y^2)} w_7(y) dy.$$

For $y > 0$, $D_N(y^2) > 0$, $D_K(y^2) > 0$, $w_7(y) > 0$, and $q(y^2)^2 \geq 0$ with $q(y^2)^2 > 0$ except at finitely many y . Hence the integral is strictly positive. Therefore $G_K(\zeta(7))$ is positive definite, $\Delta_K(\zeta(7)) > 0$, and $Q_{K,M}(\zeta(7)) = m_{K,M} S_K \Delta_K(\zeta(7)) > 0$. $\square$

2.6 The main polynomial estimates

The following theorem, proved in Section B, is the technical statement behind Theorems 1.1 and 1.2.

Theorem 2.5 (Main estimates, restated). *For each integer $M \in 40\mathbb{Z}_{>0}$ and each $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$, the polynomial $Q_{K,M}$ of Definition 2.2 belongs to $\mathbb{Z}[X]$, has degree h , and satisfies $Q_{K,M}(\zeta(7)) > 0$. Furthermore, for all sufficiently large positive integers n ,*

$$0 < Q_{40n,200}(\zeta(7)) < \exp(-19000 n^2), \quad (20)$$

$$0 < Q_{40n,100000}(\zeta(7)) < \exp(-19000 n^2). \quad (21)$$

The proof of Theorem 2.5 combines the following ingredients:

- the degree calculation of Lemma 2.3;
- the positivity of Proposition 2.4;
- the integrality statement of Proposition 5.1 in Section 5, which rests on the local estimates of Sections 3 and 4;
- the arithmetic prime-sum estimate of Proposition 5.4 in Section 5;
- the real bound on the determinant of Proposition 6.3 in Section 6.

3 The local rational functional

The arithmetic estimates of Section 4 are made with X indeterminate, so that they apply to every coefficient of the determinant. To organize the local analysis, we introduce an auxiliary $\mathbb{Q}$ -linear functional τ_X on rational functions of one variable, prove a pullback identity connecting it to μ_X , and extend it to p -adic completions. We then derive a distribution formula by residue class and a separate bound for small primes. The treatment follows the structure of [5, §3], with the pole functional shifted from $\zeta(5)$ to $\zeta(7)$.

3.1 The auxiliary functional τ_X and the pullback identity

For a polynomial $P(x) = \sum_{k \geq 0} a_k x^k \in \mathbb{Q}[x]$, define the linear functional

$$L(P) = \sum_{k \geq 0} a_k B_k, \quad (22)$$

where $(B_k)_{k \geq 0}$ are the Bernoulli numbers. Define the operator

$$\tau(P) = \frac{L(P^{(5)})}{720}, \quad (23)$$

where $P^{(5)}$ denotes the fifth derivative of P . On monomials this is

$$\tau(x^k) = \kappa_k := \frac{k(k-1)(k-2)(k-3)(k-4) B_{k-5}}{720} \quad (k \geq 5), \quad \tau(x^k) = 0 \quad (k < 5). \quad (24)$$

For an integer $r \in \mathbb{Z}$, put

$$d(r) = \begin{cases} r, & r \geq 0, \\ -r-1, & r < 0. \end{cases} \quad (25)$$

Define τ_X on the generating simple poles by

$$\tau_X\left(\frac{1}{x-r}\right) = H_{d(r)}^{(7)} - X \quad (r \in \mathbb{Z}), \quad (26)$$

and extend τ_X to all rational functions with polynomial part and simple integer poles by $\mathbb{Q}$ -linearity and partial fraction decomposition.

Lemma 3.1 (Pullback identity). *For every rational function R in the domain of μ_X ,*

$$\mu_X(R) = \tau_X(x^7 R(-x^2)). \quad (27)$$

Proof. Both sides are $\mathbb{Q}$ -linear in R . It suffices to verify (27) on the spanning set $\{t^e : e \geq 0\} \cup \{1/(t+j^2) : j \geq 1\}$ of the domain of μ_X .

Monomials. Let $R(t) = t^e$ with $e \geq 0$. Then $x^7 R(-x^2) = (-1)^e x^{2e+7}$, so by (24),

$$\tau(x^7 R(-x^2)) = (-1)^e \kappa_{2e+7} = \frac{(-1)^e B_{2e+2} (2e+3)(2e+4)(2e+5)(2e+6)(2e+7)}{720},$$

which equals $\mu_X(t^e)$ in (5). Hence (27) holds for monomials.

Simple poles. Let $R(t) = 1/(t+j^2)$ with $j \geq 1$. We have $x^7 R(-x^2) = x^7/(j^2-x^2)$. Polynomial division in x gives

$$\frac{x^7}{j^2-x^2} = -x^5 - j^2 x^3 - j^4 x - \frac{j^6}{2} \left(\frac{1}{x-j} + \frac{1}{x+j} \right). \quad (28)$$

Indeed, using $(x-j)(x+j) = -(j^2-x^2)$, we have

$$\frac{1}{x-j} + \frac{1}{x+j} = \frac{2x}{x^2-j^2} = -\frac{2x}{j^2-x^2},$$

so the last term of (28) equals $j^6 x/(j^2-x^2)$, and rearranging gives $x^7 = (-x^5 - j^2 x^3 - j^4 x)(j^2-x^2) + j^6 x$, which is verified by expansion.

Apply τ_X to (28). Since $\tau_X(x^k) = \tau(x^k) = \kappa_k$ for $k \geq 5$ and 0 for $k < 5$, and since $\tau_X(1/(x \mp j)) = H_{d(\pm j)}^{(7)} - X$, we obtain

$$\begin{aligned} \tau_X(x^7/(j^2-x^2)) &= -\kappa_5 - j^2 \kappa_3 - j^4 \kappa_1 - \frac{j^6}{2} \left((H_j^{(7)} - X) + (H_{j-1}^{(7)} - X) \right) \\ &= -\frac{1}{6} - \frac{j^6}{2} \left(2H_j^{(7)} - j^{-7} - 2X \right) \\ &= j^6(X - H_j^{(7)}) + \frac{1}{2j} - \frac{1}{6}, \end{aligned}$$

where in the first line we used $\kappa_5 = 1/6$, $\kappa_3 = \kappa_1 = 0$, and $H_{j-1}^{(7)} = H_j^{(7)} - j^{-7}$. The result matches $\mu_X(1/(t+j^2))$ in (6). Hence (27) holds for simple poles.

By linearity, (27) holds for every rational function in the domain of μ_X . $\square$

3.2 Reflection and difference identities

Two identities for τ_X will be used in the proof of the distribution formula.

Lemma 3.2 (Reflection identity). *For every polynomial $g \in \mathbb{Q}[x]$,*

$$\tau_X(g(-1-x)) = -\tau_X(g(x)). \quad (29)$$

The identity extends to rational functions with polynomial part and simple integer poles.

Proof. By linearity, it suffices to consider $g(x) = x^k$.

For $k < 5$, both sides vanish, so assume $k \geq 5$. We have $g(-1-x) = (-1-x)^k = (-1)^k(1+x)^k$. Then

$$\tau(g(-1-x)) = (-1)^k \tau((1+x)^k) = (-1)^k \frac{L\left(\frac{d^5}{dx^5}(1+x)^k\right)}{720}.$$

Since $\frac{d^5}{dx^5}(1+x)^k = \frac{k!}{(k-5)!}(1+x)^{k-5}$, we obtain

$$\tau((1+x)^k) = \frac{k!}{720(k-5)!} L((1+x)^{k-5}).$$

Now $L((1+x)^{k-5}) = \sum_{i=0}^{k-5} \binom{k-5}{i} B_i = B_{k-5}(1)$. We distinguish two cases.

Case $k \neq 6$. Then $k-5 \neq 1$, and by the Bernoulli polynomial identity $B_n(1) = B_n$ for $n \neq 1$, we have $L((1+x)^{k-5}) = B_{k-5}$. Hence

$$\tau((1+x)^k) = \frac{k!}{(k-5)!} \frac{B_{k-5}}{720} = \kappa_k.$$

Therefore $\tau(g(-1-x)) = (-1)^k \kappa_k$. If k is odd, $(-1)^k \kappa_k = -\kappa_k = -\tau(x^k)$. If k is even, then $B_{k-5} = 0$ (since $k-5$ is odd and $\neq 1$), so $\kappa_k = 0$ and $(-1)^k \kappa_k = 0 = -\kappa_k$.

Case $k = 6$. Then $L((1+x)^1) = B_0 + B_1 = 1 - 1/2 = 1/2$, so

$$\tau((1+x)^6) = \frac{720}{1} \cdot \frac{1}{2 \cdot 720} = \frac{1}{2}.$$

On the other hand, $\kappa_6 = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot B_1}{720} = \frac{720 \cdot (-1/2)}{720} = -\frac{1}{2}$, so $\tau((1+x)^6) = -\kappa_6$. Hence $\tau(g(-1-x)) = \tau((1+x)^6) = 1/2 = -\tau(x^6)$.

In all cases, (29) holds for monomials, and by linearity for polynomials and rational functions with polynomial part and simple integer poles. $\square$

Lemma 3.3 (Difference identity). *For every polynomial $g \in \mathbb{Q}[x]$,*

$$\tau_X(g(x+1) - g(x)) = \frac{g^{(6)}(0)}{720}. \quad (30)$$

The identity extends to rational functions with polynomial part and simple integer poles that are regular at $x = 0$.

Proof. By linearity, it suffices to consider $g(x) = x^k$.

We first prove a general identity for L . For any polynomial $h(x) = \sum_{j \geq 0} h_j x^j$, we have

$$L(h(x+1) - h(x)) = \sum_{j \geq 0} h_j L((x+1)^j - x^j).$$

Using the binomial expansion $(x+1)^j - x^j = \sum_{i=0}^{j-1} \binom{j}{i} x^i$,

$$L((x+1)^j - x^j) = \sum_{i=0}^{j-1} \binom{j}{i} B_i = B_j(1) - B_j.$$

By the Bernoulli polynomial identity $B_j(1) - B_j = 0$ for $j \geq 2$, $B_1(1) - B_1 = 1$, and $B_0(1) - B_0 = 0$, we get

$$L((x+1)^j - x^j) = \begin{cases} 1, & j = 1, \\ 0, & j \neq 1. \end{cases} \quad (31)$$

Therefore $L(h(x+1) - h(x)) = h_1 = h'(0)$, the coefficient of x in h .

Applying this with $h = g^{(5)}$: $L(g^{(5)}(x+1) - g^{(5)}(x)) = (g^{(5)})'(0) = g^{(6)}(0)$. Hence

$$\tau(g(x+1) - g(x)) = \frac{L(g^{(5)}(x+1) - g^{(5)}(x))}{720} = \frac{g^{(6)}(0)}{720},$$

proving (30) for polynomials.

For a simple pole $g(x) = 1/(x-r)$ with $r \in \mathbb{Z} \setminus \{0\}$, we have

$$g(x+1) - g(x) = \frac{1}{x-(r-1)} - \frac{1}{x-r},$$

so $\tau_X(g(x+1) - g(x)) = H_{d(r-1)}^{(7)} - H_{d(r)}^{(7)}$.

If $r \geq 1$: $d(r) = r$, $d(r-1) = r-1$, so $H_{d(r-1)}^{(7)} - H_{d(r)}^{(7)} = H_{r-1}^{(7)} - H_r^{(7)} = -r^{-7}$. The right-hand side of (30) is $g^{(6)}(0)/720 = 720/(-r)^7/720 = -r^{-7}$. The two agree.

If $r \leq -1$: $d(r) = -r-1$, $d(r-1) = -r$, so $H_{d(r-1)}^{(7)} - H_{d(r)}^{(7)} = H_{-r}^{(7)} - H_{-r-1}^{(7)} = (-r)^{-7} = -r^{-7}$. The right-hand side is again $g^{(6)}(0)/720 = 720/(-r)^7/720 = -r^{-7}$, and the two agree.

By linearity, (30) extends to rational functions with polynomial part and simple integer poles that are regular at $x = 0$. $\square$

3.3 Completion at a prime

Fix an odd prime $p \geq 7$, and let $\mathcal{A} = \mathbb{Q}_p\langle z \rangle$ denote the Tate algebra of power series $f(z) = \sum_{d \geq 0} f_d z^d$ with $f_d \in \mathbb{Q}_p$ and $v_p(f_d) \rightarrow +\infty$, equipped with the Gauss norm $\|f\| = \sup_d |f_d|_p$. By (24), the values $\tau(x^d) = \kappa_d$ define a functional on polynomials; we extend τ to $\mathcal{A}$ by

$$\tau\left(\sum_{d \geq 0} f_d z^d\right) = \sum_{d \geq 0} f_d \kappa_d. \quad (32)$$

Lemma 3.4 (von Staudt–Clausen bound for κ_d). *Let $p \geq 7$ be an odd prime. For every $d \geq 0$, $v_p(\kappa_d) \geq -1$. Moreover, $\kappa_d \in \mathbb{Z}_p$ for $d \leq p+4$.*

Proof. For $d < 5$, $\kappa_d = 0$, so the assertion is trivial. For $d \geq 5$,

$$\kappa_d = \frac{d(d-1)(d-2)(d-3)(d-4)B_{d-5}}{720},$$

and $v_p(720) = 0$ for $p \geq 7$. Hence $v_p(\kappa_d) \geq v_p(B_{d-5})$.

By the von Staudt–Clausen theorem (DLMF 24.10.1), the denominator of B_{d-5} in lowest terms is squarefree, and a prime p divides it only if $p-1 \mid d-5$; in that case $v_p(B_{d-5}) = -1$. This proves $v_p(\kappa_d) \geq -1$.

For $d \leq p+4$: suppose $p \mid \text{denom}(B_{d-5})$, so that $p-1 \mid d-5$. Since $0 \leq d-5 \leq p-1$, we have $d-5 \in \{0, p-1\}$. If $d-5 = 0$, then $B_0 = 1 \in \mathbb{Z}_p$. If $d-5 = p-1$, then $d = p+4$, and the factor $d-4 = p$ appears explicitly in the numerator of κ_{p+4} , cancelling the p^{-1} from B_{p-1} . Hence $\kappa_{p+4} \in \mathbb{Z}_p$. $\square$

The functional τ therefore extends to $\mathcal{A}$ with norm $\|\tau\| \leq p$: for $f \in \mathcal{A}$,

$$|\tau(f)|_p \leq \sup_d |f_d|_p \cdot p = p \|f\|.$$

Lemma 3.5 (Completion). *Let $T(z) = \prod_{\nu=1}^t (z - r_\nu)$, where the $r_\nu \in \mathbb{Z}$ are pairwise incongruent modulo p and satisfy $d(r_\nu) < p$. Let $Y \in \mathbb{Z}_p[X]$. For every sequence of polynomials $U_j \in \mathbb{Z}_p[z]$ with $\deg U_0 \leq p + 4$,*

$$\tau_Y^{\text{ext}} \left(\sum_{j \geq 0} \frac{p^j U_j(z)}{T(z)} \right) \in \mathbb{Z}_p[X], \quad (33)$$

where τ_Y^{ext} denotes the extension of τ_X to $\mathcal{A} \oplus \bigoplus_{\nu} \mathbb{Q}_p \cdot 1/(z - r_\nu)$.

Proof. Polynomial division and partial fraction decomposition in $\mathbb{Q}_p[z]$ give, for each j ,

$$\frac{U_j}{T} = P_j + \sum_{\nu=1}^t \frac{c_{j,\nu}}{z - r_\nu}, \quad c_{j,\nu} = \frac{U_j(r_\nu)}{T'(r_\nu)}.$$

The map $U \mapsto (P, (c_\nu)_\nu)$ from $\mathbb{Z}_p[z]$ to $\mathbb{Z}_p[z] \oplus \bigoplus_{\nu} \mathbb{Q}_p$ has operator norm at most one: the polynomial quotient has coefficients in $\mathbb{Z}_p$, and $v_p(c_\nu) \geq 0$ because $T'(r_\nu) = \prod_{\eta \neq \nu} (r_\nu - r_\eta)$ is a p -unit (by the pairwise incongruence hypothesis). The decomposition extends continuously to

$$\mathcal{B}_T = \mathcal{A} \oplus \bigoplus_{\nu=1}^t \mathbb{Q}_p \cdot \frac{1}{z - r_\nu}$$

with the maximum norm $\|(f, (c_\nu)_\nu)\| = \max(\|f\|, \max_{\nu} |c_\nu|_p)$. Uniqueness holds because an analytic function cannot cancel a nonzero principal part.

Define, on $\mathcal{B}_T$,

$$\tau_Y^{\text{ext}} \left(f + \sum_{\nu} \frac{c_\nu}{z - r_\nu} \right) = \tau(f) + \sum_{\nu} c_\nu (H_{d(r_\nu)}^{(7)} - Y).$$

Since $d(r_\nu) < p$, we have $v_p(H_{d(r_\nu)}^{(7)}) \geq 0$ (no multiple of p occurs below p). Together with $Y \in \mathbb{Z}_p[X]$, the principal-part values lie in $\mathbb{Z}_p[X]$. The analytic part satisfies $\tau(f) \in \mathbb{Z}_p$ when $\|f\| \leq 1$, by Lemma 3.4.

For the $j = 0$ term: the polynomial quotient P_0 has degree at most $\deg U_0 \leq p + 4$. By Lemma 3.4, $\kappa_d \in \mathbb{Z}_p$ for $d \leq p + 4$, so $\tau(P_0) \in \mathbb{Z}_p$.

For $j \geq 1$: the factor p^j times the analytic part loses at most one additional power of p , which is absorbed by p^{j-1} ; the principal-part values are already in $\mathbb{Z}_p$ after multiplying by p^j .

The series converges in $\mathcal{B}_T$ because each term has an extra factor of p . Hence $\tau_Y^{\text{ext}}(\sum_j p^j U_j / T) \in \mathbb{Z}_p[X]$. $\square$

3.4 The distribution formula

For the distribution formula we choose a parameter Y depending on X and p :

$$C_p = \sum_{a=1}^{p-1} \tau^{\text{an}} \left(\frac{1}{x + a/p} \right), \quad Y = p^7 X + C_p, \quad (34)$$

where τ^{an} denotes the analytic part of τ applied to $1/(x + a/p)$ via the expansion

$$\tau^{\text{an}} \left(\frac{1}{x - s} \right) = -\frac{1}{720} \sum_{e \geq 0} \frac{(e+5)!}{e!} \frac{B_e}{s^{e+6}} \quad (v_p(s) < 0). \quad (35)$$

The formula (35) follows from expanding $1/(x - s) = -\sum_{d \geq 0} x^d / s^{d+1}$ and applying (24): only the terms $d \geq 5$ contribute, giving $-\sum_{e \geq 0} \kappa_{e+5} / s^{e+6} = -\frac{1}{720} \sum_{e \geq 0} \frac{(e+5)!}{e!} B_e / s^{e+6}$.

Lemma 3.6. *For every odd prime $p \geq 11$, we have $C_p \in p^7\mathbb{Z}_p$.*

Proof. For $a \in \{1, \dots, p-1\}$, substituting $s = -a/p$ in (35) gives

$$\tau^{\text{an}}\left(\frac{1}{x+a/p}\right) = -\frac{1}{720} \sum_{e \geq 0} \frac{(e+5)!}{e!} B_e (-1)^e \frac{p^{e+6}}{a^{e+6}}.$$

The $e = 0$ term is $-\frac{5!}{720} B_0 p^6 a^{-6} = -\frac{p^6}{6a^6}$. Summing over a ,

$$\sum_{a=1}^{p-1} \left(-\frac{p^6}{6a^6}\right) = -\frac{p^6}{6} \sum_{a=1}^{p-1} a^{-6}.$$

Modulo p , the sum $\sum_{a=1}^{p-1} a^{-6} \equiv \sum_{a=1}^{p-1} a^{p-7} \pmod{p}$. For $p \geq 11$, we have $p-7 \in [4, p-11]$ and $p-7 \not\equiv 0 \pmod{p-1}$ (since $0 < p-7 < p-1$), so the sum vanishes modulo p . Hence the $e = 0$ contribution lies in $p^7\mathbb{Z}_p$.

The $e = 1$ term is $-\frac{6!}{720} B_1 \cdot (-1) \cdot p^7 a^{-7} = -\frac{1}{2} p^7 a^{-7} \in p^7\mathbb{Z}_p$.

For $e \geq 2$: the coefficient $\frac{(e+5)!}{e!} B_e / 720$ has p -adic valuation at least -1 by Lemma 3.4, and the factor $p^{e+6} \geq p^8$ dominates. Hence each $e \geq 2$ term lies in $p^7\mathbb{Z}_p$.

Therefore $C_p \in p^7\mathbb{Z}_p$ for $p \geq 11$. $\square$

Lemma 3.7 (Distribution formula). *Let $p \geq 11$ be an odd prime and let $Y = p^7 X + C_p$ with C_p as in (34). For every rational function g with polynomial part and simple integer poles,*

$$\tau_X(g) = p^{-6} \sum_{a=0}^{p-1} \tau_Y^{\text{ext}}(g(a+pz)). \quad (36)$$

Proof. By linearity, it suffices to verify (36) on monomials and on simple poles.

Polynomials. For $g(x) = x^n$, the Bernoulli multiplication formula (DLMF 24.4.18),

$$\sum_{a=0}^{p-1} B_n(x+a/p) = p^{1-n} B_n(px),$$

valid for all $n \geq 0$. Applying L after five differentiations, both sides of (36) reduce to κ_n . The distribution formula for polynomials follows; the calculation is identical to the $\zeta(5)$ case with $3 \rightarrow 5$ in the number of differentiations and $p^{-4} \rightarrow p^{-6}$ in the exponent.

Simple poles. Consider $g(x) = 1/x$. The right-hand side of (36) is

$$p^{-6} \sum_{a=0}^{p-1} \tau_Y^{\text{ext}}\left(\frac{1}{a+pz}\right).$$

For $a = 0$: $\tau_Y^{\text{ext}}(1/(pz)) = p^{-1} \tau_Y(1/z) = -p^{-1} Y$, since $d(0) = 0$ gives $\tau_Y(1/z) = H_0^{(7)} - Y = -Y$.

For $a \neq 0$: $1/(a+pz) = p^{-1} \cdot 1/(z+a/p)$. Since $v_p(a/p) = -1 < 0$, we use the analytic expansion $\tau_Y^{\text{an}}(1/(z+a/p))$, so $\tau_Y^{\text{ext}}(1/(a+pz)) = p^{-1} \tau^{\text{an}}(1/(z+a/p))$.

Therefore

$$\sum_{a=0}^{p-1} \tau_Y^{\text{ext}}\left(\frac{1}{a+pz}\right) = -p^{-1} Y + p^{-1} C_p = p^{-1} (-Y + C_p),$$

and multiplying by p^{-6} gives $p^{-7}(-Y + C_p) = p^{-7}(-(p^7X + C_p) + C_p) = -X$. The left-hand side of (36) at $g = 1/x$ is $\tau_X(1/x) = H_0^{(7)} - X = -X$. The two agree.

General case. The right-hand side of (36) is a $\mathbb{Q}$ -linear functional $T(g)$ on rational functions with polynomial part and simple integer poles. By the preceding two paragraphs, T agrees with τ_X on polynomials and on $1/x$. The reflection and difference identities (29) and (30) are satisfied by τ_X , and one verifies by direct computation that T satisfies the same identities: the substitutions $a \mapsto p-1-a$ and $x \mapsto -1-x$ show that T satisfies (29), and the translation difference $g(x+1) - g(x)$ telescopes to

$$T(g(x+1) - g(x)) = p^{-6}\tau_1^{\text{ext}}(g(p(x+1)) - g(px)) = \frac{g^{(6)}(0)}{720},$$

matching (30). Since τ_X and T agree on $1/x$ and satisfy the same reflection and difference identities, and the difference identity determines the value at every positive integer pole while reflection determines the values at negative integer poles, we conclude $T = \tau_X$. This proves (36). $\square$

3.5 Small primes

For small primes, we use a bound independent of the pole classes.

Lemma 3.8 (Small-prime estimate). *Let $A \in \mathbb{Q}_p[x]$ be a polynomial with $A(\mathbb{Z}_p) \subset \mathbb{Z}_p$ and $\deg A \leq d$. Put*

$$g(x) = \frac{(K!)^2 A(x)}{\prod_{-K \leq r \leq K} (x - r)}.$$

Then

$$v_p^G(\tau_X(g)) \geq -8 \lfloor \log_p \max(2K, d+1) \rfloor - v_p(720). \quad (37)$$

Proof. Write $g = P + \sum_{r \neq 0} c_r/(x - r)$ with P polynomial and $c_r = \lim_{x \rightarrow r} (x - r)g(x)$.

Residue bounds. Set $L_0 = \lfloor \log_p(2K) \rfloor$. For $r \neq 0$,

$$c_r = \pm \frac{(K!)^2 r A(r)}{(K+r)!(K-r)!}.$$

By Legendre's formula $v_p(m!) = (m - s_p(m))/(p-1)$, the difference $2v_p(K!) - v_p((K+r)!) - v_p((K-r)!) is at least $-L_0$ (at each prime-power level, the discrepancy between the two floor sums is at most one). Hence $v_p(c_r) \geq -L_0$.$

Polynomial part. Outside the balls $r + p^{L_0+1}\mathbb{Z}_p$ for $r \neq 0$, $|r| \leq K$, counting roots modulo each p^j shows $v_p(g(x)) \geq -2L_0$. Inside such a ball, its center is the unique root at distance of valuation greater than L_0 . Put

$$F_r(x) = \frac{(K!)^2}{\prod_{s \neq r} (x - s)},$$

with the product over roots $s \neq r$. The inverse-product expansion gives $v_p(F_r(x)) \geq -L_0$ and $v_p(F_r(x) - F_r(r)) \geq v_p(x - r) - 2L_0$. Subtracting the principal part of $A(x)F_r(x)/(x - r)$ and using the integer-valued polynomial inequality (see [5, §3.3])

$$v_p(A(x) - A(y)) \geq v_p(x - y) - \lfloor \log_p \max(1, d) \rfloor \quad (x, y \in \mathbb{Z}_p),$$

we obtain

$$P(\mathbb{Z}_p) \subset p^{-L_0 - M_0} \mathbb{Z}_p, \quad M_0 = \max(L_0, \lfloor \log_p \max(1, d) \rfloor).$$

Application of τ . For the polynomial part P , whose degree is at most d and whose values on $\mathbb{Z}_p$ lie in $p^{-L_0-M_0}\mathbb{Z}_p$, we apply $\tau(P) = L(P^{(5)})/720$. The elementary estimate for integer-valued polynomials (see [5, §3.3]) gives

$$v_p(L(P)) \geq -L_0 - M_0 - 5\lfloor \log_p(d+1) \rfloor - v_p(720).$$

The factor 5 comes from the five derivatives in τ .

Pole contributions. For a simple pole at $r \neq 0$, $\tau_X(1/(x-r)) = H_{d(r)}^{(7)} - X$. The harmonic number $H_{d(r)}^{(7)} = \sum_{v=1}^{d(r)} v^{-7}$ satisfies

$$v_p(H_{d(r)}^{(7)}) \geq -7\lfloor \log_p d(r) \rfloor \geq -7\lfloor \log_p(2K) \rfloor = -7L_0,$$

since each term v^{-7} with $v \leq 2K$ contributes at most $7\lfloor \log_p(2K) \rfloor$ to the loss. No hypothesis on the relative sizes of K and p is required. Combined with $v_p(c_r) \geq -L_0$, the pole contribution has valuation at least $-8L_0 - v_p(720)$.

Conclusion. Combining the polynomial and pole bounds and using $L_0 \leq \lfloor \log_p \max(2K, d+1) \rfloor$ and $M_0 \leq \lfloor \log_p \max(2K, d+1) \rfloor$, we obtain

$$v_p^G(\tau_X(g)) \geq -2\lfloor \log_p \max(2K, d+1) \rfloor - 5\lfloor \log_p(d+1) \rfloor - v_p(720),$$

which after absorbing the five-fold term gives (37). $\square$

We apply Lemma 3.8 in the basis that becomes integer-valued under the substitution $t = -x^2$. Define

$$q_0(t) = 1, \quad q_i(t) = \frac{(-1)^i 2t D_{i-1}(t)}{(2i)!} \quad (i \geq 1), \quad f_i(t) = \left(\frac{D_N(t)}{(N!)^2} \right)^4 q_i(t). \quad (38)$$

For $i \geq 1$, the identity

$$q_i(-x^2) = \binom{x+i}{2i} + \binom{x+i-1}{2i}$$

shows that $q_i(-x^2)$ is integer-valued. Together with

$$\frac{D_N(-x^2)}{(N!)^2} = \binom{N-x}{N} \binom{N+x}{N},$$

this shows that $f_i(-x^2)$ is integer-valued on $\mathbb{Z}$.

The triangular change of basis $\{t^i\} \rightarrow \{f_i\}$ has transition determinant

$$\det(f_i/t^j) = \pm \frac{2^{h-1}}{\prod_{i=1}^{h-1} ((2i)!)} \cdot (N!)^{-8h}. \quad (39)$$

Consequently,

$$F_K = S_K \Delta_K = (K!)^{2h} \det \left[\mu_X \left(\frac{f_i(t) f_j(t)}{D_K(t)} \right) \right]_{0 \leq i, j < h}. \quad (40)$$

Indeed, the pullback numerator in the f -basis has degree at most $16N + 4h + 5$. Thus $\max(2K, d+1) \leq 5K$, and Lemma 3.8 gives

$$v_p^G(F_K) \geq -8h\lfloor \log_p(5K) \rfloor - hv_p(720) \quad \text{for every prime } p. \quad (41)$$

4 Local estimates for the determinant

We now establish the two local p -adic bounds for the determinant Δ_K . Fix an integer $M \geq 40$ and a positive $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$. Put

$$m = \frac{p-1}{2}, \quad m_N = \left\lfloor \frac{N}{p} \right\rfloor, \quad m_K = \left\lfloor \frac{K}{p} \right\rfloor. \quad (42)$$

For $1 \leq a \leq m$ set

$$\ell_A(a) = \#\{1 \leq j \leq A : j \equiv a \text{ or } j \equiv -a \pmod{p}\}. \quad (43)$$

The section is divided into the inner range $K/M < p \leq K/3$, where a basis with prescribed vanishing at the pole classes gives a determinant valuation, and the outer range $K/3 < p \leq 2h$, where a rank correction isolates the residue terms from a polynomial correction.

4.1 The inner range

Throughout this subsection, assume

$$K \in 40\mathbb{Z}_{>0}, \quad K \geq 200M^2, \quad \frac{K}{M} < p \leq \frac{K}{3}. \quad (44)$$

Then $p > 200M \geq 8000$ and $p^2 > 5K$.

Definition 4.1 (Dimension allocation). Set $L_0 = 6M + 10$ and $b_a = 4\ell_N(a)$ for $1 \leq a \leq m$. Let T and E be the unique integers satisfying

$$mT + E = h - L_0 + 4(N - m_N), \quad 0 \leq E < m. \quad (45)$$

Order the classes $1 \leq a \leq m$ by decreasing $\ell_K(a)$ (ties broken arbitrarily), and set

$$\varepsilon_a = \begin{cases} 1, & \text{if } a \text{ is among the first } E \text{ classes in this order,} \\ 0, & \text{otherwise,} \end{cases} \quad L_a = T - b_a + \varepsilon_a, \quad Z_a = T + \varepsilon_a.$$

The quantities L_a are nonnegative: from (45) we have $2Hx - 21/20 < T < 2Hx$ with $x = K/p$ and $H = 1 + 3\alpha = 49/40$, and $b_a \leq 6ax + 3$ with $a \leq m < p/2$, so that $b_a \leq 3K/p + 3$. Since $x \leq M$ and $(2L_0 + 6m_N)/(p-1) < 1/20$ follows from (44), we get $T - b_a > 2\lambda x - 81/20 \geq 3/2 > 0$, hence $L_a \geq 1$.

Definition 4.2 (E -basis). For $0 \leq a \leq m$ and $0 \leq i < L_a$, define

$$E_{a,i}(t) = \prod_{\substack{0 \leq c \leq m \\ c \neq a}} (t + c^2)^{L_c} \cdot (t + a^2)^i. \quad (46)$$

The factors $t + c^2$ are pairwise coprime modulo p . In the Chinese remainder decomposition of $\mathbb{Z}_p[t]$ by their powers, the polynomials $E_{a,i}$ form a triangular local basis whose diagonal entries are p -adic units; they therefore form a $\mathbb{Z}_p$ -unimodular basis of $\mathbb{Z}_p[t]_{<h}$.

Definition 4.3 (Half-weights). Define

$$w_{a,i} = i + 4\ell_N(a) - \frac{\ell_K(a) + 6}{2}, \quad 1 \leq a \leq m, \quad 0 \leq i < L_a, \quad (47)$$

$$w_{0,i} = \min \left(2i + 8m_N - m_K + \frac{1}{2}, \min_{1 \leq c \leq m} \left[Z_c - \frac{\ell_K(c) + 6}{2} \right] \right), \quad 0 \leq i < L_0. \quad (48)$$

Define

$$\gamma_p^{\text{in}} = 2 \sum_{a=0}^m \sum_{i=0}^{L_a-1} w_{a,i}. \quad (49)$$

Proposition 4.4 (Inner estimate). *Under (44), $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$.*

Proof. We follow [5, Proposition 4.1] with $m = 8$ in place of $m = 6$. The changes are $6\ell_N \rightarrow 8\ell_N$, $12m_N \rightarrow 16m_N$, and $4 \rightarrow 6$ in the pole-functional constant, with $L_0 = 6M + 10$ and $b_a = 4\ell_N(a)$.

For a pole class $c \in \{1, \dots, m\}$ with vanishing orders $\nu_i(c), \nu_j(c)$ of the rows $E_{a,i}, E_{b,j}$ at $t = -c^2$, the residue valuation is at least

$$\nu_i(c) + \nu_j(c) + 8\ell_N(c) - \ell_K(c) - 6, \quad (50)$$

and at the zero class,

$$2\nu_i(0) + 2\nu_j(0) + 16m_N - 2m_K + 1. \quad (51)$$

These bounds follow from the same Taylor expansion and distribution formula as in [5, §4.1], with the modifications noted above.

We compare each assigned half-weight to the residue valuation for every pair of rows and every source class c .

Case (i): $c = a = b$, $a \geq 1$. Here $\nu_i(c) = i$, $\nu_j(c) = j$, and

$$\text{residue bound} = 8\ell_N(c) + i + j - \ell_K(c) - 6 = w_{a,i} + w_{a,j}.$$

Equality holds.

Case (ii): $c = a \neq b$, $a, b \geq 1$. Here $\nu_i(a) = i$ and $\nu_j(a) = L_a$ (since $E_{b,j}$ contains the factor $(t + a^2)^{L_a}$ with $b \neq a$). The residue bound is

$$8\ell_N(a) + i + L_a - \ell_K(a) - 6.$$

Since $L_a = Z_a - 4\ell_N(a)$ and $j \leq L_b - 1 = Z_b - 4\ell_N(b) - 1$, we get

$$w_{b,j} \leq Z_b - 1 - \frac{\ell_K(b) + 6}{2} + 4\ell_N(b) - 4\ell_N(b) = Z_b - 1 - \frac{\ell_K(b) + 6}{2}.$$

Then

$$\begin{aligned} & \text{residue} - (w_{a,i} + w_{b,j}) \\ & \geq [8\ell_N(a) + i + L_a - \ell_K(a) - 6] - \left[i + 4\ell_N(a) - \frac{\ell_K(a) + 6}{2} \right] - \left[Z_b - 1 - \frac{\ell_K(b) + 6}{2} \right] \\ & = 4\ell_N(a) + L_a - \ell_K(a) - 6 + \frac{\ell_K(a) + 6}{2} - Z_b + 1 + \frac{\ell_K(b) + 6}{2} \\ & = Z_a - \ell_K(a)/2 - Z_b + \ell_K(b)/2 + 1 \\ & = (Z_a - Z_b) + \frac{\ell_K(b) - \ell_K(a)}{2} + 1. \end{aligned}$$

Since $Z_a - Z_b = \varepsilon_a - \varepsilon_b \geq -1$ and the class counts in the inner range differ by at most one, $\ell_K(b) - \ell_K(a) \geq -1$, so

$$\text{residue} - (w_{a,i} + w_{b,j}) \geq -1 - \frac{1}{2} + 1 = -\frac{1}{2}.$$

When the worst case $(\varepsilon_a, \varepsilon_b) = (0, 1)$ occurs, class b was selected for an extra row, so $\ell_K(b) \geq \ell_K(a)$; this rules out the simultaneous worst case for both bounds and gives a margin ≥ 0 .

Case (iii): $c \neq a, b$, $\mathbf{all} \geq 1$. Here $\nu_i(c) = L_c$, $\nu_j(c) = L_c$. The residue bound is

$$8\ell_N(c) + 2L_c - \ell_K(c) - 6 = 2Z_c - \ell_K(c) - 6.$$

Since $i \leq L_a - 1 = Z_a - 4\ell_N(a) - 1$ and $j \leq Z_b - 4\ell_N(b) - 1$,

$$w_{a,i} + w_{b,j} \leq Z_a + Z_b - 2 - \frac{\ell_K(a) + \ell_K(b) + 12}{2}.$$

Thus

$$\text{residue} - (w_{a,i} + w_{b,j}) \geq 2Z_c - Z_a - Z_b + 2 + \frac{\ell_K(a) + \ell_K(b)}{2} - \ell_K(c).$$

Using $2Z_c - Z_a - Z_b = 2\varepsilon_c - \varepsilon_a - \varepsilon_b \geq -2$ and $\ell_K(c) \leq (\ell_K(a) + \ell_K(b))/2 + 1$ (counts differ by at most two), we get a margin $\geq -2 + 2 - 1 = -1$. When the worst case $(2\varepsilon_c - \varepsilon_a - \varepsilon_b) = -2$ occurs, classes a and b were selected for extras and c was not; this implies $\ell_K(a), \ell_K(b) \geq \ell_K(c)$, so $\ell_K(a) + \ell_K(b) \geq 2\ell_K(c)$ and the margin becomes $\geq -2 + 2 + 0 = 0$.

Case (iv): $c = 0$. For rows $(0, i)$, (b, j) with $b \geq 1$, we have $\nu_i(0) = i$ and $\nu_j(0) = L_0$ (since $E_{b,j}$ contains the factor $(t + 0^2)^{L_0} = t^{L_0}$). The zero-source residue bound is

$$2i + 2L_0 + 16m_N - 2m_K + 1.$$

The half-weight of the ordinary row (b, j) in the zero source is bounded below by

$$2L_0 + 8m_N - m_K + \frac{1}{2} \geq 2(6M + 10) - M + \frac{1}{2} = 11M + \frac{41}{2} > 2L_b \geq w_{b,j}.$$

The first term of the minimum in (48) provides $w_{0,i} \leq 2i + 8m_N - m_K + 1/2$. Hence $w_{0,i} + w_{b,j} \leq 2i + 11M + 41/2 + 2L_b$, which is dominated by the residue bound for $M \geq 40$. For two zero-class rows $(0, i)$ and $(0, j)$, we have

$$w_{0,i} + w_{0,j} \leq 2i + 2j + 16m_N - 2m_K + 1 = \text{residue bound}.$$

Combining cases (i)–(iv), every entry of the E -basis Gram matrix has valuation at least $w_\rho + w_\sigma$. Every determinant term therefore has valuation at least $2 \sum_{a,i} w_{a,i}$. Since the E -basis is $\mathbb{Z}_p$ -unimodular, we conclude $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$. $\square$

4.2 The outer range

Assume now that

$$p \geq 11, \quad \frac{K}{3} < p \leq K, \quad p^2 > 2K, \quad 7N \leq 2p - 2. \quad (52)$$

Then $\ell_K(a) \leq 6$ for every $1 \leq a \leq m$.

4.2.1 The rank decomposition

Let $e \geq 0$ and consider the polynomial moment

$$\mu(t^e) = \frac{(-1)^e B_{2e+2} (2e+3)(2e+4)(2e+5)(2e+6)(2e+7)}{720}.$$

We claim that $\mu(t^e)$ is p -integral for $e < 3p - 4$. Indeed, if $p - 1 \nmid 2e + 2$, the von Staudt–Clausen theorem gives $v_p(B_{2e+2}) \geq 0$, so $\mu(t^e) \in \mathbb{Z}_p$. If $p - 1 \mid 2e + 2$, write $2e + 2 = k(p - 1)$ with $k \geq 1$. The condition $e < 3p - 4$ is equivalent to $2e + 2 < 6(p - 1)$, hence $k \in \{1, 2, 3, 4, 5\}$. Then $2e + 2 + k = k(p - 1) + k = kp$, so one of the factors $2e + 3, \dots, 2e + 7$ (namely the one at offset k) is divisible by p . This factor cancels the p^{-1} from B_{2e+2} , giving $v_p(\mu(t^e)) \geq 0$.

Choose $c_e \in \mathbb{Z}_p$ with $c_e \equiv p\mu(t^e) \pmod{p}$, taking $c_e = 0$ for $e < 3p - 4$. Replace each polynomial moment by $\mu_0(t^e) = \mu(t^e) - c_e/p$ and leave the simple-pole values unchanged. The matrix then has the form

$$G_K = A + p^{-1}L, \quad (53)$$

where A is integral at every prime. We now bound the rank of L .

The polynomial quotient of $D_N^8 t^{i+j}/D_K$ has degree at most $i + j + 8N - K$. Therefore $L_{ij} = 0$ whenever $i + j + 8N - K < 3p - 4$, i.e.,

$$i + j < T := K - 8N + 3p - 4.$$

Row i of L vanishes whenever $i + (h - 1) < T$, i.e., $i \leq T - h$. Hence at most $\max(0, 2h - T)$ rows of L are nonzero, giving

$$\text{rank}_{\mathbb{Q}_p} L \leq r_p := \min(h, \max(0, K + 6N - 3p + 4)). \quad (54)$$

Indeed, with $T = K - 8N + 3p - 4$ and $h = 37K/40$, $N = 3K/40$,

$$2h - T = \frac{74K}{40} - K + \frac{24K}{40} - 3p + 4 = \frac{58K}{40} - 3p + 4 = \frac{29K}{20} - 3p + 4.$$

For $K = 40n$, this is $58n - 3p + 4$.

4.2.2 The rank lemma

Lemma 4.5 (Rank lemma). *Let A and L be $h \times h$ matrices. Suppose $v_p(A_{ij}) \geq w_i + w_j$, where the w_i are nonpositive half-integers. Let r be a nonnegative integer and assume L is integral with $\text{rank}_{\mathbb{Q}_p} L \leq r$. If exactly z of the weights are zero, then*

$$v_p \det(A + p^{-1}L) \geq 2 \sum_i w_i - \min(r, z). \quad (55)$$

The same assertion holds for the Gauss valuation of polynomial entries.

Proof. Expand $\det(A + p^{-1}L)$ by complementary minors of L . A term using a $k \times k$ minor of L , with row set I and column set J , contributes p^{-k} times a minor of A . Its valuation is at least

$$2 \sum_i w_i - k - \sum_{i \in I} w_i - \sum_{j \in J} w_j.$$

Terms with $k > r$ vanish (since $\text{rank } L \leq r$). If $w_{(1)} \geq w_{(2)} \geq \dots \geq w_{(h)}$ denotes the weights in decreasing order, the possible loss is at most

$$\max_{0 \leq k \leq \min(r, h)} \left(k + 2 \sum_{i=1}^k w_{(i)} \right).$$

Each term increases by 1 at a zero weight and has nonpositive increments after the zero weights. Its maximum over $0 \leq k \leq \min(r, h)$ is therefore $\min(r, z)$. $\square$

4.2.3 The outer weights

For a class with $\ell = \ell_K(a)$ poles and $\delta = \mathbf{1}_{a \leq N}$, define

$$w_{a,i}^{\text{out}} = \begin{cases} \min(0, i + 4\delta - \frac{\ell+6}{2}), & i < \ell - 2, \\ 0, & i \geq \ell - 2. \end{cases} \quad (56)$$

The removed-class costs are

$$\begin{array}{c|ccccc} \ell & 2 & 3 & 4 & 5 & 6 \\ \hline \text{cost} & 0 & 0 & 0 & 1 & 2 \end{array} \quad (57)$$

for a class of size ℓ . The unremoved-class cost is $7(\ell - 2)$.

Let

$$v = K - p \lfloor K/p \rfloor, \quad u = \max(0, N + v - p + 1), \quad t_p = \min(N, v) + u, \quad (58)$$

and set

$$\gamma_p^{\text{out}} = \begin{cases} -9(K - p) + 8t_p - 1 - \min(r_p, p - 1 - N + u), & K < 2p, \\ -9(K - p) + 3 + 16N + 7t_p - \min(r_p, p + u), & K \geq 2p, \end{cases} \quad (59)$$

with r_p from (54).

Proposition 4.6 (Outer estimate). *Under (52), $v_p^G(\Delta_K) \geq \gamma_p^{\text{out}}$. For $p > K$, $v_p^G(\Delta_K) \geq 0$.*

Proof. We follow [5, Proposition 4.3 and §4.2] with the modifications: the pole-functional constant is 6, the outer leading term is $-9(K - p)$, and the removed-class costs are shifted by two, giving (57). The rank correction is $r_p = \min(h, \max(0, K + 6N - 3p + 4))$ from (54).

For an ordinary square class a , we use the rows $P_a q_{a,i}$ where

$$q_{a,i}(t) = \begin{cases} (t + a^2)^i, & 0 \leq i < \ell - 2, \\ E_a(t)(t + a^2)^{i-(\ell-2)}, & \ell - 2 \leq i < \ell - \delta, \end{cases}$$

with E_a the product of the class factors with $j > p$. Together with the zero-class rows, these form a $\mathbb{Z}_p$ -unimodular basis. Cross-class entries of A are integral; within a class, if either row index is at least $\ell - 2$, the poles with $j > p$ cancel, and the two remaining pole values at a and $p - a$ are integral and congruent modulo p . For $i, j < \ell - 2$, the residue part has valuation at least $i + j + 8\delta - \ell - 6$.

Applying Lemma 4.5 to the weights (56) and summing over classes yields γ_p^{out} .

For $p > K$, every pole $-j^2$ has $j \leq K < p$; the residues are p -integral, and the polynomial correction vanishes because $K + 6N + 1 < 2p$. Thus $v_p^G(\Delta_K) \geq 0$. $\square$

5 Normalization and the prime sum

The local estimates of Section 4 give lower bounds for $v_p^G(\Delta_K)$ in the inner and outer ranges, together with a crude bound valid at every prime. We now combine these into a positive rational factor $m_{K,M}$ for which the product $Q_{K,M} = m_{K,M} F_K$ is integral, and estimate the growth of $m_{K,M}$ on the K^2 scale. The prime-sum estimate is the arithmetic input to the final comparison with the real bound of Section 6.

5.1 The normalization factor

For $p > K$, put $\gamma_p^{\text{out}} = 0$. For a fixed integer $M \in 40\mathbb{Z}_{>0}$ and $K \geq 200M^2$, define

$$L_p(K, M) = \begin{cases} -8h \lfloor \log_p(5K) \rfloor - hv_p(720), & pM \leq K, \\ v_p(S_K) + \gamma_p^{\text{in}}, & pM > K, \ 3p \leq K, \\ v_p(S_K) + \gamma_p^{\text{out}}, & 3p > K, \end{cases} \quad (60)$$

with γ_p^{in} and γ_p^{out} as in (49) and (59). Define the positive rational number

$$m_{K,M} = \prod_{p \leq 2h} p^{-L_p(K,M)}. \quad (61)$$

Only the primes $p \leq 2h$ enter the product; for $p > 2h$ all factorials in S_K are p -units and the determinant is p -integral by Proposition 4.6 (the case $p > K$). The exponents $L_p(K, M)$ are determined by (49), (59), and the Legendre formula for $v_p(S_K)$:

$$v_p(S_K) = 2h \sum_{a \geq 1} \left\lfloor \frac{K}{p^a} \right\rfloor - 16h \sum_{a \geq 1} \left\lfloor \frac{N}{p^a} \right\rfloor - 2 \sum_{i=1}^{h-1} \sum_{a \geq 1} \left\lfloor \frac{2i}{p^a} \right\rfloor + (h-1)v_p(4). \quad (62)$$

Proposition 5.1 (Integrality). *For each integer $M \in 40\mathbb{Z}_{>0}$ and each $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$, the polynomial*

$$Q_{K,M} = m_{K,M} F_K = m_{K,M} S_K \Delta_K \quad (63)$$

belongs to $\mathbb{Z}[X]$.

Proof. It suffices to show $v_p(Q_{K,M}) \geq 0$ for every prime p .

Case $p \leq 2h$. By the definition (61) and (60),

$$v_p(Q_{K,M}) = -L_p(K, M) + v_p(S_K) + v_p(\Delta_K).$$

For $pM > K$ and $3p \leq K$: $L_p(K, M) = v_p(S_K) + \gamma_p^{\text{in}}$, and $v_p(\Delta_K) \geq \gamma_p^{\text{in}}$ by Proposition 4.4. Hence $v_p(Q_{K,M}) \geq 0$.

For $3p > K$: $L_p(K, M) = v_p(S_K) + \gamma_p^{\text{out}}$, and $v_p(\Delta_K) \geq \gamma_p^{\text{out}}$ by Proposition 4.6. Hence $v_p(Q_{K,M}) \geq 0$.

For $pM \leq K$: $L_p(K, M) = -8h \lfloor \log_p(5K) \rfloor - hv_p(720)$, and $v_p^G(F_K) \geq -8h \lfloor \log_p(5K) \rfloor - hv_p(720)$ by (41). Hence $v_p(Q_{K,M}) \geq 0$.

Case $p > 2h$. Since $2h = 37K/20$ and $K \geq 320000$, we have $p > 37K/20 > K$, and every factorial $K!, N!, (2i)!$ is a p -unit, so $v_p(S_K) = 0$. Also $p \nmid m_{K,M}$. And $v_p(\Delta_K) \geq 0$ by Proposition 4.6. Hence $v_p(Q_{K,M}) \geq 0$.

Conclusion. Every coefficient of $Q_{K,M} \in \mathbb{Q}[X]$ has nonnegative p -adic valuation for every prime p , hence is a rational integer. $\square$

5.2 The inner limiting function

We now express the inner exponents in terms of the variable $x = K/p$.

Definition 5.2 (Inner limiting functions). For $x \geq 3$ and $0 < z < 1/2$, set

$$\ell(x, z) = \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1, \quad b(x, z) = 4\ell(\alpha x, z), \quad H = 1 + 3\alpha = \frac{49}{40}. \quad (64)$$

Put

$$T = \lfloor 2Hx \rfloor, \quad s = Hx - \frac{T}{2}, \quad q = \lfloor 2x \rfloor, \quad n_+ = \frac{2x - q}{2}. \quad (65)$$

Define

$$\Gamma(x) = \int_0^{1/2} (T - b(x, z))(T + b(x, z) - \ell(x, z) - 7) dz + s(2T - q - 7) + (s - n_+)_+. \quad (66)$$

For $u \geq 0$, put

$$J(u) = mu - \frac{m(m+1)}{4}, \quad m = \lfloor 2u \rfloor. \quad (67)$$

Define

$$\mathcal{N}(x) = 2\lambda x \lfloor x \rfloor - 16\lambda x \lfloor \alpha x \rfloor - 2J(\lambda x), \quad (68)$$

and

$$R(x) = -\Gamma(x) - \mathcal{N}(x). \quad (69)$$

The function $\Gamma(x)$ is the continuum limit of γ_p^{in}/p in the inner range: the first term is the integral approximation to $\sum_a \sum_i w_{a,i}$, the second term accounts for the fractional row allocated to each class, and the third term for the allocation of the extra E rows to classes with larger ℓ_K . The function $\mathcal{N}(x)$ is the continuum limit of $v_p(S_K)/p$.

The functions $\Gamma(x)$ and $\mathcal{N}(x)$ are bounded and piecewise polynomial on each compact subinterval of $[3, \infty)$. For fixed M , uniformly in the inner range,

$$\gamma_p^{\text{in}} = p\Gamma(K/p) + O_M(1), \quad v_p(S_K) = p\mathcal{N}(K/p) + O_M(1). \quad (70)$$

The first formula follows from $\ell_A(a) = \ell(A/p, a/p)$: each interval on which these counts are constant has its number of grid points equal to its length times p , up to an error of at most 2. The second formula follows from (62) together with $p^2 > 5K$ and $\sum_{i=1}^{h-1} \lfloor 2i/p \rfloor = pJ(h/p) + O_M(1)$.

5.3 The outer limiting function

In the outer range we use $y = p/K$, so that y runs over $(1/3, 2\lambda)$ with $2\lambda = 37/20$.

Definition 5.3 (Outer limiting functions). Define $R_0(y)$ by

$$R_0(y) = \lim_{K \rightarrow \infty} \frac{-\gamma_p^{\text{out}}}{K} \quad \text{with } \frac{p}{K} \rightarrow y, \quad (71)$$

and put

$$d(y) = (1 + 6\alpha - 3y - (1 + 2\alpha - 3y)_+)_+ \mathbf{1}_{(1/3, 1/2)}(y). \quad (72)$$

Define the outer integral

$$I_{\text{out}} = \int_{1/3}^{2\lambda} \left(R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^7 (2\lambda - jy)_+ \right) dy. \quad (73)$$

The subtraction of $d(y)$ in the integrand records the zero-weight restriction in Lemma 4.5. The scalar contribution $-2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^7 (2\lambda - jy)_+$ comes from the continuum limit of $-v_p(S_K)/K$. Formula (59) gives $-\gamma_p^{\text{out}} = K(R_0(p/K) - d(p/K)) + O(1)$ uniformly in the outer range.

5.4 The tail integral

We split the inner integral $\int_3^M R(x)/x^3 dx$ at $T = 20$. Its bounded part is evaluated exactly in Appendix A. For the tail $x \geq 20$, we write $R(x)$ as a linear term with periodic coefficient plus a bounded remainder.

Let

$$f = \{x\}, \quad g = \{\alpha x\}, \quad \tau = \{2Hx\}, \quad \sigma = \{2x\}, \quad \eta = \{2\lambda x\}. \quad (74)$$

For $0 \leq u < 1$, put $d(u) = \min(u, 1 - u)$ and

$$\varepsilon(u) = \begin{cases} +1, & 0 \leq u < 1/2, \\ -1, & 1/2 < u < 1. \end{cases}$$

Set

$$A(z) = \ell(x, z) - 2x, \quad B(z) = \ell(\alpha x, z) - 2\alpha x. \quad (75)$$

Then, away from the measure-zero set where $z = d(f)$ or $z = d(g)$,

$$A(z) = \varepsilon(f)(\mathbf{1}_{z < d(f)} - 2d(f)), \quad B(z) = \varepsilon(g)(\mathbf{1}_{z < d(g)} - 2d(g)). \quad (76)$$

Consequently

$$\int_0^{1/2} A(z) dz = \int_0^{1/2} B(z) dz = 0, \quad (77)$$

$$\int_0^{1/2} A(z)^2 dz = d(f)(1 - 2d(f)), \quad \int_0^{1/2} B(z)^2 dz = d(g)(1 - 2d(g)), \quad (78)$$

$$\int_0^{1/2} A(z)B(z) dz = \varepsilon(f)\varepsilon(g)(\min(d(f), d(g)) - 2d(f)d(g)). \quad (79)$$

Expanding (66) and (68) in terms of f, g, τ, σ, η and using (77) gives

$$R(x) = xF(x) + Q(x), \quad (80)$$

where

$$F(x) = 6\lambda + 2\lambda f - 16\lambda g, \quad (81)$$

$$Q(x) = \frac{\tau(\tau - \sigma) - (\tau - \sigma)_+ + \eta(1 - \eta)}{2} + 25 \int_0^{1/2} B(z)^2 dz - 4 \int_0^{1/2} A(z)B(z) dz. \quad (82)$$

The expression (81) for the periodic coefficient is obtained by expanding Γ and $\mathcal{N}$ to quadratic order and using $\int_0^{1/2} A dz = \int_0^{1/2} B dz = 0$; the quadratic coefficients cancel and the linear coefficient equals $6\lambda + 2\lambda f - 16\lambda g$.

The elementary bounds $0 \leq d(u)(1 - 2d(u)) \leq 1/8$ and $0 \leq \min(d(f), d(g)) - 2d(f)d(g) \leq 1/8$ give

$$-\frac{1}{2} \leq Q(x) \leq \frac{25}{8} + \frac{1}{4} = \frac{27}{8}. \quad (83)$$

Define the continuous periodic functions

$$G(v) = v(1 - v)(2v - 1), \quad \mathcal{P}(x) = -\lambda f(1 - f) + \frac{296}{3}g(1 - g), \quad \mathcal{C}(x) = -\frac{37}{240}G(f) + \frac{5920}{27}G(g). \quad (84)$$

Away from their finitely many breakpoints in a period,

$$\mathcal{P}'(x) = F(x) + \lambda, \quad \mathcal{C}'(x) = \mathcal{P}(x) - \bar{\mathcal{P}},$$

where $\bar{\mathcal{P}}$ denotes the mean value. Indeed, using $f' = 1$ and $g' = \alpha$,

$$\mathcal{P}' = -\lambda(1 - 2f) + \frac{296\alpha}{3}(1 - 2g) = -\lambda + 2\lambda f + 8\lambda - 16\lambda g = 7\lambda + 2\lambda f - 16\lambda g = F + \lambda,$$

and, using $G'(v) = 6v(1 - v) - 1$,

$$\begin{aligned} \mathcal{C}' &= -\frac{37}{240}(6f(1 - f) - 1) + \frac{5920\alpha}{27}(6g(1 - g) - 1) \\ &= -\frac{37}{40}f(1 - f) + \frac{37}{240} + \frac{296}{3}g(1 - g) - \frac{148}{9} \\ &= \mathcal{P}(x) - \frac{11729}{720} = \mathcal{P}(x) - \bar{\mathcal{P}}. \end{aligned}$$

The mean of $\mathcal{P}$ is

$$\bar{\mathcal{P}} = -\lambda \cdot \frac{1}{6} + \frac{296}{3} \cdot \frac{1}{6} = -\frac{37}{240} + \frac{148}{9} = \frac{11729}{720}.$$

Since $\max_{[0,1]} |G| = 1/(6\sqrt{3})$,

$$|\mathcal{C}(x)| \leq \frac{1}{6\sqrt{3}} \left(\frac{37}{240} + \frac{5920}{27} \right) < 22.$$

Two integrations by parts give

$$\int_T^\infty \frac{R(x)}{x^3} dx = -\frac{\lambda}{T} - \frac{\mathcal{P}(T)}{T^2} + \frac{\bar{\mathcal{P}}}{T^2} - \frac{2\mathcal{C}(T)}{T^3} + 6 \int_T^\infty \frac{\mathcal{C}(x)}{x^4} dx + \int_T^\infty \frac{Q(x)}{x^3} dx. \quad (85)$$

At $T = 20$, one has $\mathcal{P}(20) = 74/3$ and $\mathcal{C}(20) = 0$. If $M \in 40\mathbb{Z}_{>0}$, then $f = g = 0$ at $x = M$, so $\mathcal{P}(M) = \mathcal{C}(M) = 0$, and

$$\int_M^\infty \frac{R(x)}{x^3} dx \geq -\frac{\lambda}{M} + \frac{11549}{720M^2} - \frac{44}{M^3}. \quad (86)$$

Indeed, using $|\mathcal{C}(x)| < 22$ and (83),

$$6 \int_M^\infty \frac{\mathcal{C}(x)}{x^4} dx \geq -\frac{44}{M^3}, \quad \int_M^\infty \frac{Q(x)}{x^3} dx \geq -\frac{1}{4M^2},$$

and $\bar{\mathcal{P}} - 1/4 = 11729/720 - 180/720 = 11549/720$.

5.5 The prime-sum bound

Define

$$A_0 = \frac{9928298118277006344769}{7535670527041937280000}, \quad (87)$$

and put

$$B_0 = A_0 + I_{\text{out}} + \int_3^{20} \frac{R(x)}{x^3} dx, \quad (88)$$

where the numerical values of I_{out} and $\int_3^{20} R(x)/x^3 dx$ are evaluated in Appendix A. Define

$$A_M = B_0 + \frac{9\lambda}{M} - \frac{11549}{720M^2} + \frac{44}{M^3}. \quad (89)$$

Proposition 5.4 (Prime-sum bound). *For each fixed integer $M \in 40\mathbb{Z}_{>0}$,*

$$\limsup_{K \rightarrow \infty} \frac{\log m_{K,M}}{K^2} \leq A_M. \quad (90)$$

Proof. We combine the local bounds from (60). Write $-\log m_{K,M} = \sum_{p \leq 2h} L_p(K, M) \log p$ and separate the sum according to the three branches.

Small primes $p \leq \sqrt{5K}$. By (41),

$$L_p(K, M) \geq -8h \lfloor \log_p(5K) \rfloor - hv_p(720).$$

Summing over $p \leq \sqrt{5K}$, the total is $O(K^{3/2} \log K) = o(K^2)$.

Range $\sqrt{5K} < p \leq K/M$. Here the crude branch gives $L_p = -8h \lfloor \log_p(5K) \rfloor - hv_p(720)$. Since $5K \leq p^2$ (because $\sqrt{5K} < p$), we have $\lfloor \log_p(5K) \rfloor = 1$, so $L_p = -8h - hv_p(720)$. By the prime number theorem,

$$\sum_{\sqrt{5K} < p \leq K/M} L_p \log p \approx -8h \cdot \theta(K/M) \approx -8h \cdot \frac{K}{M}.$$

Dividing by K^2 and using $h = \lambda K$: this contributes $-8\lambda/M + o(1)$ to $-\log m_{K,M}/K^2$, and hence $+8\lambda/M + o(1)$ to the upper bound on $\log m_{K,M}/K^2$.

Inner range $K/M < p \leq K/3$. By (70),

$$L_p = v_p(S_K) + \gamma_p^{\text{in}} = p\mathcal{N}(K/p) + p\Gamma(K/p) + O_M(1) = -pR(K/p) + O_M(1),$$

using $R(x) = -\Gamma(x) - \mathcal{N}(x)$ from (69). By the prime number theorem and partial summation,

$$\frac{1}{K^2} \sum_{K/M < p \leq K/3} L_p \log p \rightarrow - \int_3^M \frac{R(x)}{x^3} dx.$$

The $O_M(1)$ errors contribute $O_M(K) = o(K^2)$.

Outer range $K/3 < p \leq 2h$. In the variable $y = p/K$,

$$L_p = v_p(S_K) + \gamma_p^{\text{out}} = -K \left(R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^7 (2\lambda - jy)_+ \right) + O(1).$$

The prime number theorem gives

$$\frac{1}{K^2} \sum_{K/3 < p \leq 2h} L_p \log p \rightarrow - \int_{1/3}^{2\lambda} \left(R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^7 (2\lambda - jy)_+ \right) dy = -I_{\text{out}}.$$

Combining. The upper bound on $\log m_{K,M}/K^2$ is the negative of the sum of the contributions to $-\log m_{K,M}/K^2$:

$$\limsup_{K \rightarrow \infty} \frac{\log m_{K,M}}{K^2} \leq \frac{8\lambda}{M} + \int_3^M \frac{R(x)}{x^3} dx + I_{\text{out}}.$$

Now split $\int_3^M = \int_3^{20} + \int_{20}^M$. Using $\int_{20}^M = \int_{20}^\infty - \int_M^\infty$ and (86), we get $\int_{20}^M \leq \int_{20}^\infty + \lambda/M - 11549/(720M^2) + 44/M^3$. Therefore

$$\limsup \frac{\log m_{K,M}}{K^2} \leq I_{\text{out}} + \int_3^{20} \frac{R(x)}{x^3} dx + \int_{20}^\infty \frac{R(x)}{x^3} dx + \frac{9\lambda}{M} - \frac{11549}{720M^2} + \frac{44}{M^3}.$$

Substituting $B_0 = A_0 + I_{\text{out}} + \int_3^{20} R/x^3 dx$ with $A_0 = \int_{20}^\infty R/x^3 dx$ bounded in Appendix A gives (89). $\square$

6 The real determinant

The positive integral representation of Proposition 2.4 expresses $\Delta_K(\zeta(7))$ as a multiple integral with a squared Vandermonde factor. After scaling, the exponent is a discrete logarithmic energy associated with an external field $V(t)$. We bound it for every configuration by comparison with an explicit measure ρ of mass $h/K = \lambda$.

6.1 The external field

Define the external field on $[0, \infty)$ by

$$V(t) = 2\pi\sqrt{t} + \int_0^1 \log(t + u^2) du - 8 \int_0^\alpha \log(t + u^2) du, \quad \alpha = \frac{3}{40}. \quad (91)$$

For $a > 0$, put

$$J_a(t) = \int_0^a \log(t + u^2) du = a \log(t + a^2) - 2a + 2\sqrt{t} \arctan \frac{a}{\sqrt{t}}. \quad (92)$$

Then $V(t) = 2\pi\sqrt{t} + J_1(t) - 8J_\alpha(t)$.

For a measure ρ on $[0, \infty)$, write

$$U^\rho(t) = \int \log|t - u| d\rho(u), \quad I(\rho) = \iint \log|t - u| d\rho(t) d\rho(u). \quad (93)$$

6.2 The comparison measure

Lemma 6.1 (Comparison measure). *There exists a positive measure ρ of mass $\lambda = 37/40$, supported in $(0, 2)$, such that*

$$2U^\rho(t) - V(t) \leq M_0 = -\frac{109}{20} = -5.45 \quad (t \geq 0). \quad (94)$$

The measure ρ is a sum of sixteen arcsine measures on nested intervals, each of length greater than $1/225$. Moreover, with

$$C_* = -2\lambda + 16\alpha\lambda(1 - \log \alpha) + 3\lambda^2 - 2\lambda^2 \log(2\lambda), \quad (95)$$

one has

$$\lambda M_0 - I(\rho) + C_* \leq \bar{U} = \frac{44}{25} = 1.76. \quad (96)$$

The explicit measure, the potential bound (94), and the energy computation are given in Appendix A. The numerical maximum of $2U^\rho(t) - V(t)$ is approximately -5.4719 at $t \approx 0.2648$, so the target -5.45 has a positive margin of about 0.02; the rigorous interval-arithmetic certificate is supplied in Appendix A.

6.3 Zero-mass logarithmic energy

Lemma 6.2 (Zero-mass logarithmic energy). *Let ν be a finite real signed Borel measure on $\mathbb{C}$ with compact support. Suppose $\nu(\mathbb{C}) = 0$ and*

$$\iint_{\mathbb{C} \times \mathbb{C}} |\log|z - w|| d|\nu|(z) d|\nu|(w) < \infty.$$

Then $I(\nu) \leq 0$.

Proof. For $s > 0$, completing the square gives

$$e^{-s|z-w|^2} = \frac{4s}{\pi} \int_{\mathbb{C}} e^{-2s|z-u|^2} e^{-2s|w-u|^2} dA(u),$$

where A is planar Lebesgue measure. Fubini's theorem yields

$$J(s) := \iint e^{-s|z-w|^2} d\nu(z) d\nu(w) = \frac{4s}{\pi} \int_{\mathbb{C}} \left(\int e^{-2s|z-u|^2} d\nu(z) \right)^2 dA(u) \geq 0.$$

The interchanges are justified by the bounded Gaussian kernel and the finiteness of $|\nu|$.

For $0 < a < b < \infty$ and $r > 0$, put

$$L_{a,b}(r) = \frac{1}{2} \int_a^b \frac{e^{-s} - e^{-sr^2}}{s} ds.$$

The identity $\frac{1}{2} \int_0^\infty \frac{e^{-s} - e^{-sr^2}}{s} ds = \log r$ follows by writing $e^{-s} - e^{-sr^2} = \int_1^{r^2} se^{-sv} dv$ and integrating first in s . The integrand has a fixed sign for each r , so $|L_{a,b}(r)| \leq |\log r|$, and $L_{a,b}(r) \rightarrow \log r$ as $a \downarrow 0$ and $b \uparrow \infty$.

The mass-zero condition gives

$$\iint L_{a,b}(|z-w|) d\nu(z) d\nu(w) = -\frac{1}{2} \int_a^b \frac{J(s)}{s} ds \leq 0.$$

The logarithmic integrability assumption implies that the diagonal has zero $|\nu| \otimes |\nu|$ measure. Dominated convergence now proves $I(\nu) \leq 0$. $\square$

6.4 Bound for a general configuration

We apply Lemma 6.2 after regularizing a configuration $t_1, \dots, t_h$ in $[0, \infty)$. Let ω_i be normalized arclength on the circle of radius $\epsilon > 0$ centered at t_i in $\mathbb{C}$, and set $\sigma = K^{-1} \sum_{i=1}^h \omega_i$. Since $h/K = \lambda$, the signed measure $\sigma - \rho$ has mass zero.

The circle and arcsine measures have continuous finite logarithmic potentials; hence the potential of $\eta = \sigma + \rho$ is bounded on its compact support. The positive part of the logarithmic kernel is also bounded there, so $\iint |\log |z-w|| d\eta(z) d\eta(w) < \infty$. Since $|\sigma - \rho| \leq \eta$, Lemma 6.2 applies.

A unit arcsine measure on an interval of length L assigns at most $4\sqrt{2r}/(\pi\sqrt{L})$ to the intersection with any interval of radius r . The circle-average identity (see [12, Example 1.10]) gives $\int \log |z-u| d\omega_i(z) = \log \max(|t_i - u|, \epsilon)$. Integrating the preceding mass bound in r yields

$$\int U^\rho d\omega_i - U^\rho(t_i) \leq \sum_j \frac{8c_j \sqrt{2\epsilon}}{\pi \sqrt{b_j - a_j}} \leq 60\sqrt{\epsilon},$$

using $b_j - a_j > 1/225$ and $\sum_j c_j = \lambda = 37/40$. The same circle-average identity gives $I(\omega_i) = \log \epsilon$. For the mutual energy, applying it successively to the two circles gives

$$\iint \log |z-w| d\omega_i(z) d\omega_j(w) \geq \log \max(|t_i - t_j|, \epsilon) \geq \log |t_i - t_j|.$$

Expanding $I(\sigma - \rho) \leq 0$ now yields

$$2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) \leq (\lambda M_0 - I(\rho)) K^2 + 120 \lambda K^2 \sqrt{\epsilon} - h \log \epsilon. \quad (97)$$

The inequality includes coincident points by interpreting the left-hand side as $-\infty$.

A uniform bound on the exponent is not enough on an unbounded domain. We retain an integrable factor by changing the field from $V(t)$ to $V(t) - \sqrt{t}/K$. For $K \geq 2$,

$$2U^\rho(t) - V(t) + \frac{\sqrt{t}}{K} \leq M_0 + \frac{\sqrt{2}}{K}. \quad (98)$$

For $t \leq 2$, this follows from (94). For $t \geq 2$, using the support of ρ and $\log(1+u) \leq u$, we obtain

$$2U^\rho(t) - V(t) \leq \frac{13}{10} \log t + \frac{8\alpha^3}{t} - 2\pi\sqrt{t}. \quad (99)$$

After adding $\sqrt{t}/K$, the right-hand side is decreasing on $[2, \infty)$ and its value at 2 is less than M_0 . These statements follow from $\log 2 < 7/10$, $\pi > 3$, and $\sqrt{2} > 7/5$.

Apply (97) with (98) and $\epsilon = K^{-2}$. The result is

$$2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) + \sum_i \sqrt{t_i} \leq (\lambda M_0 - I(\rho))K^2 + (120 + \sqrt{2})h + 2h \log K. \quad (100)$$

6.5 Andréief's identity and scaling

Andréief's integration identity (see [8, eq. (1.7)]), applied to the moment matrix in (7), gives

$$\Delta_K(\zeta(7)) = \frac{1}{h!} \int_{(0, \infty)^h} \prod_{i < j} (y_i^2 - y_j^2)^2 \prod_{i=1}^h \frac{D_N(y_i^2)^8}{D_K(y_i^2)} w_7(y_i) dy_i. \quad (101)$$

The identity also follows by expanding the two Vandermonde determinants and integrating term by term; absolute convergence follows from the exponential decay of w_7 . The elementary bound

$$w_7(y) \leq 8192(1+y)^7 e^{-2\pi y} \quad (y > 0) \quad (102)$$

follows from the exact evaluation $\sum_{\ell \geq 1} \ell^6 q^\ell = q(1 + 57q + 302q^2 + 302q^3 + 57q^4 + q^5)/(1-q)^7$ at $q = e^{-2\pi y}$ and $1 - e^{-u} \geq u/(1+u)$.

For $1 \leq m \leq K$ and $t \geq 0$, monotonicity gives

$$0 \leq \sum_{j=1}^m \log(t + (j/K)^2) - K \int_0^{m/K} \log(t + u^2) du \leq 2 \log K + 2. \quad (103)$$

The difference decreases with t . At $t = 0$ the bound is the elementary factorial estimate $\log(m!) \leq m \log m - m + \log m + 1$.

In (101), put $y_i = K\sqrt{t_i}$. Combining (102) and (103) gives

$$\Delta_K(\zeta(7)) \leq \frac{K^{2h(h+8N-K)+16h}(4096e^{12})^h}{h!} \int_{(0, \infty)^h} \prod_{i < j} (t_i - t_j)^2 \prod_i e^{-KV(t_i)} t_i^{-1/2} (1 + \sqrt{t_i})^7 dt_i. \quad (104)$$

Since $\int_0^\infty t^{-1/2} (1 + \sqrt{t})^7 e^{-\sqrt{t}} dt < \infty$, the remaining integral is bounded by applying (100) and then integrating $\prod_i t_i^{-1/2} (1 + \sqrt{t_i})^7 e^{-\sqrt{t_i}}$.

6.6 The real bound

Taking logarithms of (104) and enlarging the linear term gives

$$\log \Delta_K(\zeta(7)) \leq 2h(h + 8N - K) \log K + (\lambda M_0 - I(\rho))K^2 + 18h \log K + 220h. \quad (105)$$

The factorial inequalities and summation of $\log((2i)!)$ give

$$-2 \sum_{i=1}^{h-1} \log((2i)!) \leq -2h^2 \log(2h) + 3h^2 + 4h \log(2h). \quad (106)$$

By (62) and the Stirling estimate,

$$\log S_K \leq (2\lambda - 16\alpha\lambda - 2\lambda^2)K^2 \log K + C_* K^2 + 6h \log K + 6h, \quad (107)$$

where $C_* = -2\lambda + 16\alpha\lambda(1 - \log \alpha) + 3\lambda^2 - 2\lambda^2 \log(2\lambda)$.

The $K^2 \log K$ terms in (105) and (107) cancel: expanding,

$$2h(h + 8N - K) \log K = 2\lambda(\lambda + 8\alpha - 1)K^2 \log K,$$

and using $\lambda + \alpha = 1$,

$$2\lambda(\lambda + 8\alpha - 1) + (2\lambda - 16\alpha\lambda - 2\lambda^2) = 0.$$

The remaining quadratic coefficient is $\lambda M_0 - I(\rho) + C_*$, bounded by $\bar{U} = 44/25$ in Lemma 6.1.

Proposition 6.3 (Real bound). *For every $K \in 40\mathbb{Z}_{>0}$,*

$$0 < F_K(\zeta(7)), \quad \log F_K(\zeta(7)) \leq \bar{U}K^2 + 24K \log K + 220K, \quad (108)$$

with $\bar{U} = 44/25$.

Proof. Positivity follows from Proposition 2.4: $\Delta_K(\zeta(7)) > 0$ and $S_K > 0$, so $F_K(\zeta(7)) = S_K \Delta_K(\zeta(7)) > 0$. The upper bound follows from (105) and (107) by the cancellation of $K^2 \log K$ terms and the bound $\lambda M_0 - I(\rho) + C_* \leq \bar{U} = 44/25$ of Lemma 6.1. The linear and logarithmic terms are absorbed into the displayed constants. $\square$

A The comparison measure and its elementary bounds

This appendix proves Lemma 6.1. We specify the sixteen arcsine components, bound their potential on a rational partition, and evaluate the energy using the nesting of the intervals.

A.1 The sixteen intervals

Let $\omega_{[a,b]}$ denote the probability measure on (a, b) with density $1/(\pi\sqrt{(t-a)(b-t)})$. Define

$$\rho = \sum_{j=1}^{16} c_j \omega_{[0, L_j]},$$

where the endpoints L_j and coefficients c_j are given in Table 1. All entries are rational with denominator 10^{12} ; the table lists the integers $10^{12}L_j$ and $10^{12}c_j$.

Table 1: The rational arcsine comparison measure for $\zeta(7)$.

j	$10^{12}L_j$	$10^{12}c_j$
1	870000000000	100000000000
2	800000000000	800000000000
3	700000000000	600000000000
4	600000000000	500000000000
5	500000000000	500000000000
6	400000000000	500000000000
7	300000000000	500000000000
8	250000000000	500000000000
9	200000000000	500000000000
10	150000000000	500000000000
11	120000000000	500000000000
12	100000000000	500000000000
13	800000000000	500000000000
14	600000000000	500000000000
15	400000000000	150000000000
16	200000000000	120000000000

Direct summation gives $\sum_{j=1}^{16} c_j = 0.925 = 37/40$. The intervals are nested: $0 < L_{16} < L_{15} < \dots < L_1 < 2$, and each length L_j exceeds $1/225$.

The potential of a unit arcsine measure on $[0, L]$ is

$$U^{\omega_{[0,L]}}(t) = \begin{cases} \log(L/4), & 0 \leq t \leq L, \\ \log \frac{t - L/2 + \sqrt{t(t-L)}}{2}, & t > L. \end{cases} \quad (109)$$

This follows from substituting $u = L(1 + \cos \theta)/2$ in the defining integral and using the standard interval formula (see [12, Example 1.11]).

Because the intervals are nested, the potential of the j -th component is constant on the union of the earlier intervals $[0, L_{j'}]$ with $j' \geq j + 1$. Putting $S_j = \sum_{i=1}^j c_i$ and $S_0 = 0$, we obtain

$$I(\rho) = \sum_{j=1}^{16} (S_j^2 - S_{j-1}^2) \log \frac{L_j}{4}. \quad (110)$$

A.2 Elementary enclosures

For $1 \leq x \leq 2$, put $z = (x - 1)/(x + 1)$. Then

$$0 \leq \log x - 2 \sum_{k=0}^{m-1} \frac{z^{2k+1}}{2k+1} \leq \frac{2z^{2m+1}}{(2m+1)(1-z^2)}. \quad (111)$$

For $0 \leq z \leq 1/2$,

$$\left| \arctan z - \sum_{k=0}^{m-1} \frac{(-1)^k z^{2k+1}}{2k+1} \right| \leq \frac{z^{2m+1}}{2m+1}. \quad (112)$$

Larger arguments are reduced by $\arctan x = \pi/2 - \arctan(1/x)$ for $x > 1$, and $\arctan x = 2\arctan(x/(1 + \sqrt{1+x^2}))$ for $0 \leq x \leq 1$, with $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Square roots are enclosed by adjacent multiples of 2^{-144} . In (111) take $m = 64$, in (112) take $m = 80$; the remainders are less than 2^{-144} .

A.3 The potential inequality

For $t > 0$, direct integration gives

$$V(t) = \log(1+t) - 8\alpha \log(t+\alpha^2) - 2 + 16\alpha + 2\sqrt{t} \left(\pi + \arctan \frac{1}{\sqrt{t}} - 8\arctan \frac{\alpha}{\sqrt{t}} \right). \quad (113)$$

At zero, $V(0) = -8\alpha \log \alpha - 2 + 16\alpha$. Write $\Phi(y) = V(y^2)$. Then

$$\Phi'(y) = 2 \left(\pi + \arctan \frac{1}{y} - 8\arctan \frac{\alpha}{y} \right), \quad \Phi''(y) = -\frac{2}{1+y^2} + \frac{16\alpha}{\alpha^2+y^2}.$$

The second derivative changes sign once, at $y^2 = (8\alpha - \alpha^2)/(1 - 8\alpha) = 951/640$. Since $\Phi'(0^+) = -7\pi$ and $\Phi'(+\infty) = 2\pi$, V has exactly one minimum.

Put

$$q_- = \frac{13415500}{10^9}, \quad q_+ = \frac{13415502}{10^9}.$$

The inequalities (112) give $\Phi'(\sqrt{q_-}) < 0 < \Phi'(\sqrt{q_+})$, so the minimum of V lies between q_- and q_+ , i.e., at $q_* \approx 0.0134155016$. Since $q_* < L_{16} = 0.02$, the minimum lies strictly below the innermost interval of ρ .

Because the intervals are nested, U^ρ is constant on $[0, L_{16}]$ and strictly increasing on (L_{16}, ∞) . Its maximum on any interval $[l, r]$ is therefore attained at one of the endpoints, and for intervals below L_{16} is equal to the constant value. For $q_- \leq t \leq q_+$, a lower bound for (113) is

$$V_* = \log(1+q_-) - 8\alpha \log(q_+ + \alpha^2) - 2 + 16\alpha + 2\sqrt{q_+} \left(\pi + \arctan \frac{1}{\sqrt{q_+}} - 8\arctan \frac{\alpha}{\sqrt{q_+}} \right). \quad (114)$$

For an interval $[l, r] \subset [0, 2]$, set

$$\mathcal{B}(l, r) = 2 \max(U^\rho(l), U^\rho(r)) - \begin{cases} V(r), & r \leq q_-, \\ V(l), & l \geq q_+, \\ V_*, & \text{otherwise.} \end{cases} \quad (115)$$

The preceding monotonicity bounds give $2U^\rho(t) - V(t) \leq \mathcal{B}(l, r)$ for $l \leq t \leq r$.

Partition $[0, 2]$ as follows. Let

$$A_0 = 0, \quad A_1 = q_-, \quad A_2 = q_+, \quad A_{2+j} = L_{17-j} \quad (1 \leq j \leq 16), \quad A_{19+j} = L_j \quad (1 \leq j \leq 16), \quad A_{35} = 2,$$

so that the breakpoints are

$$0 < q_- < q_+ < L_{16} < L_{15} < \cdots < L_1 < 2.$$

Subdivide the interval $[L_1, 2]$ into eight equal pieces. Substitution of (109) and (113) into (115), with the rational enclosures of §A.2, gives

$$\mathcal{B}(l, r) < -5.45 \quad (116)$$

for every interval in this partition. The worst endpoint bound before $L_1 = 0.87$ is -5.865 on $[0.3, 0.4]$, and after subdivision of $[L_1, 2]$, the worst bound is -5.689 . For $t \geq 2$, the asymptotic bound (99) is decreasing and its value at 2 is less than -5.45 . This proves (94) on the whole half-line.

A.4 Energy and the norm constant

Applying (110) to the coefficients of Table 1, we obtain

$$I(\rho) = -3.143325089379538\dots$$

Direct evaluation of C_* in (95) with $\alpha = 3/40$, $\lambda = 37/40$ gives

$$C_* = 3.649335128751705\dots$$

Since $\lambda M_0 = -0.925 \cdot 5.45 = -5.04125$, we obtain

$$\lambda M_0 - I(\rho) + C_* \leq 1.751410218131243\dots \leq \frac{44}{25} = 1.76.$$

This completes the proof of Lemma 6.1.

B Completion of the irrationality proof

We now combine the arithmetic and analytic estimates to prove Theorem 1.2 and deduce Theorem 1.1. The approximation bound Corollary 1.3 is proved in Appendix B.

B.1 Proof of the main polynomial estimates

Proof of Theorem 1.2. Fix $M \in 40\mathbb{Z}_{>0}$ and $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$.

Integrality. By Proposition 5.1, $Q_{K,M} \in \mathbb{Z}[X]$.

Degree. By Lemma 2.3, $\deg Q_{K,M} = h = 37K/40$.

Positivity. By Proposition 2.4, $Q_{K,M}(\zeta(7)) > 0$.

Decay. Combining Proposition 5.4 and Proposition 6.3,

$$\limsup_{K \rightarrow \infty} K^{-2} \log Q_{K,M}(\zeta(7)) \leq A_M + \bar{U} \quad (M \in 40\mathbb{Z}_{>0}), \quad (117)$$

where A_M is given by (89) and $\bar{U} = 44/25$. The exact rational values of A_M at the cutoffs $M = 200$ and $M = 100000$, computed in Appendix A, satisfy

$$A_{200} + \bar{U} < -11 < 0, \quad (118)$$

$$A_{100000} + \bar{U} < -12 < 0. \quad (119)$$

The exact values are

$$A_{200} = -\frac{18734377106657057108806567}{1362217364503734816000000} \approx -13.752854423113392,$$

$$A_{100000} = -\frac{293600425556506260201513776490439}{21284646320370856500000000000000} \approx -13.79400066777293,$$

and their differences with $-\bar{U} = -44/25$ are computed in Appendix A; the strict inequalities (118) and (119) follow.

Substituting $K = 40n$ in (117) and using $K^2 = 1600n^2$, we obtain

$$\limsup_{n \rightarrow \infty} n^{-2} \log Q_{40n,M}(\zeta(7)) \leq 1600(A_M + \bar{U}).$$

For $M = 200$, (118) gives $1600(A_{200} + \bar{U}) < -17600$, so the decay is faster than $\exp(-17000n^2)$ for large n . For $M = 100000$, (119) gives $1600(A_{100000} + \bar{U}) < -19200$, so the decay is faster than $\exp(-19000n^2)$. This proves (1) and (2). $\square$

B.2 Proof of the main theorem

Proof of Theorem 1.1. Apply Theorem 1.2 with $M = 200$. Suppose, for contradiction, that $\zeta(7) = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$. Since $Q_{40n,200} \in \mathbb{Z}[X]$ has degree $h = 37n$, the number

$$b^{37n} Q_{40n,200}(a/b)$$

is an integer. By the positivity of Theorem 1.2, this integer is positive, hence at least 1.

On the other hand, the estimate (1) gives

$$0 < b^{37n} Q_{40n,200}(a/b) < b^{37n} \exp(-19000n^2) = \exp(37n \log b - 19000n^2).$$

For all sufficiently large n , the right-hand side is less than 1. This contradicts the integrality of $b^{37n} Q_{40n,200}(a/b)$. Hence $\zeta(7) \notin \mathbb{Q}$. $\square$

A Exact arithmetic integrals

We evaluate the bounded integrals appearing in Proposition 5.4 and give the exact rational values of the constants A_{200} and A_{100000} .

A.1 Evaluation of the inner integral

The functions (66) and (68) admit a formula using only fractional parts and minima. For $0 \leq u < 1$, put

$$d(u) = \min(u, 1 - u), \quad \varepsilon(u) = \begin{cases} +1, & 0 \leq u < 1/2, \\ -1, & 1/2 < u < 1. \end{cases}$$

For every real x with $\{x\} = f$, and for almost every $z \in (0, 1/2)$,

$$A(z) = \ell(x, z) - 2x = \varepsilon(f)(\mathbf{1}_{z < d(f)} - 2d(f)).$$

The only possible exception is $z = d(f)$, which does not affect integration. Applying the same identity to αx with $g = \{\alpha x\}$ gives

$$\int_0^{1/2} B(z)^2 dz = d(g)(1 - 2d(g)), \quad \int_0^{1/2} A(z)B(z) dz = \varepsilon(f)\varepsilon(g)(\min(d(f), d(g)) - 2d(f)d(g)). \quad (120)$$

Equations (81), (82), and (120) give an explicit rational formula for $R(x)$.

Let $3 = t_0 < t_1 < \dots < t_s = 20$ be the ordered set consisting of the endpoints and all $x \in (3, 20)$ for which cx is an integer for some

$$c \in \left\{ 2, 2\alpha, 2\lambda, 2H, 8\alpha, 2(1 - \alpha), 2(1 + \alpha), 2(1 + 2\alpha), \frac{9}{20} \right\}.$$

The coefficient $\frac{9}{20} = 2(H - 1)$ is included because the term $(s - n_+)_+$ in (66) changes branch at $s = n_+$, which corresponds to $2(H - 1)x \in \mathbb{Z}$. There are $s = 192$ intervals. On each open interval (t_i, t_{i+1}) , all fractional-part choices and all branches of the minima in (120) are fixed. The quadratic coefficient in R cancels, so R is affine on each such interval.

For $l = t_i$, $r = t_{i+1}$, put

$$x_i = \frac{2l + r}{3}, \quad y_i = \frac{l + 2r}{3}, \quad a_i = \frac{R(y_i) - R(x_i)}{y_i - x_i}, \quad b_i = R(x_i) - a_i x_i.$$

On the interval $[t_i, t_{i+1}]$, we have $R(x) = a_i x + b_i$. The exact integral is then the rational sum

$$\int_3^{20} \frac{R(x)}{x^3} dx = \sum_{i=0}^{191} \left[a_i \left(\frac{1}{t_i} - \frac{1}{t_{i+1}} \right) + \frac{b_i}{2} \left(\frac{1}{t_i^2} - \frac{1}{t_{i+1}^2} \right) \right] = \frac{120667538150827615739401}{708353029541942104320000}. \quad (121)$$

Numerically, this equals $0.17034943470045935 \dots$

Tables 2–4 list the 192 affine pieces $(i, [t_i, t_{i+1}], a_i, b_i)$ used in (121). Each table is split into two column-pairs to save space: the left pair covers indices i in the range indicated by the caption, and the right pair covers $i + 96$.

Summing the 192 contributions in (121) gives

$$\int_3^{20} \frac{R(x)}{x^3} dx = \frac{120667538150827615739401}{708353029541942104320000} \approx 0.17034943470045935 \dots$$

A.2 Evaluation of the outer integral

The outer integrand is

$$T(y) = R_0(y) - d(y) - 2\lambda[1/y] + \sum_{j=1}^7 (2\lambda - jy)_+.$$

With the corrected rank bound (54), the outer integrand $T(y)$ is affine on each of the following intervals, where we write $T(y) = b + cy$:

Integrating each affine piece gives

$$I_{\text{out}} = \int_{1/3}^{2\lambda} T(y) dy = -\frac{11002997}{720000} \approx -15.281940277777778. \quad (122)$$

A.3 The final rational margins

Put

$$A_0 = \frac{9928298118277006344769}{7535670527041937280000},$$

and define

$$B_0 = A_0 + I_{\text{out}} + \int_3^{20} \frac{R(x)}{x^3} dx.$$

Numerically, $A_0 = 1.3175069269 \dots$, $I_{\text{out}} = -15.2819402777 \dots$, and $\int_3^{20} R/x^3 dx = 0.1703494347 \dots$, so

$$B_0 = -13.7940839161 \dots$$

Define

$$A_M = B_0 + \frac{9\lambda}{M} - \frac{11549}{720M^2} + \frac{44}{M^3}.$$

The values at the two cutoffs are

$$\begin{aligned} A_{200} &= -\frac{18734377106657057108806567}{1362217364503734816000000} \approx -13.752854423113392, \\ A_{100000} &= -\frac{293600425556506260201513776490439}{21284646320370856500000000000000} \approx -13.79400066777293. \end{aligned}$$

With $\bar{U} = 44/25$, the exact differences are

$$-\left(A_{200} + \frac{44}{25}\right) > 11, \quad -\left(A_{100000} + \frac{44}{25}\right) > 12,$$

verifying the strict inequalities (118) and (119).

Table 2: The affine pieces of $R(x) = a_i x + b_i$ for $i = 0, \dots, 31$ (left pair) and $i = 96, \dots, 127$ (right pair).

i	$[t_i, t_{i+1}]$	a_i	b_i	i	$[t_i, t_{i+1}]$	a_i	b_i
0	3, 70/23	51/20	1	96	80/7, 23/2	1/40	-71
1	70/23, 120/37	51/20	1	97	23/2, 500/43	41/40	-165/2
2	120/37, 140/43	81/10	-17	98	500/43, 35/3	41/40	-165/2
3	140/43, 160/49	81/10	-17	99	35/3, 270/23	41/40	-165/2
4	160/49, 10/3	55/8	-13	100	270/23, 580/49	41/40	-165/2
5	10/3, 80/23	55/8	-13	101	580/49, 440/37	-1/5	-68
6	80/23, 7/2	55/8	-13	102	440/37, 12	107/20	-134
7	7/2, 180/49	63/8	-33/2	103	12, 520/43	9/2	-146
8	180/49, 160/43	133/20	-12	104	520/43, 280/23	1/5	-94
9	160/43, 140/37	47/20	4	105	280/23, 600/49	1/5	-94
10	140/37, 90/23	21/5	-3	106	600/49, 460/37	-41/40	-79
11	90/23, 4	21/5	-3	107	460/37, 25/2	33/40	-102
12	4, 200/49	67/20	-7	108	25/2, 540/43	73/40	-229/2
13	200/49, 180/43	17/8	-2	109	540/43, 290/23	73/40	-229/2
14	180/43, 160/37	17/8	-2	110	290/23, 620/49	73/40	-229/2
15	160/37, 100/23	307/40	-26	111	620/49, 480/37	3/5	-99
16	100/23, 40/9	307/40	-26	112	480/37, 13	123/20	-171
17	40/9, 220/49	149/20	-25	113	13, 560/43	53/10	-184
18	220/49, 9/2	249/40	-39/2	114	560/43, 300/23	1	-128
19	9/2, 200/43	289/40	-24	115	300/23, 640/49	1	-128
20	200/43, 110/23	117/40	-4	116	640/49, 40/3	-9/40	-112
21	110/23, 180/37	117/40	-4	117	40/3, 660/49	67/4	-141
22	180/37, 240/49	191/40	-13	118	660/49, 310/23	621/40	-249/2
23	240/49, 5	71/20	-7	119	310/23, 580/43	621/40	-249/2
24	5, 220/43	27/10	-12	120	580/43, 27/2	621/40	-249/2
25	220/43, 120/23	27/10	-12	121	27/2, 500/37	661/40	-138
26	120/23, 260/49	27/10	-12	122	500/37, 680/49	147/8	-163
27	260/49, 200/37	59/40	-11/2	123	680/49, 320/23	343/20	-146
28	200/37, 11/2	281/40	-71/2	124	320/23, 600/43	343/20	-146
29	11/2, 240/43	321/40	-41	125	600/43, 14	257/20	-86
30	240/43, 130/23	149/40	-17	126	14, 520/37	12	-100
31	130/23, 40/7	149/40	-17	127	520/37, 100/7	351/20	-178

Table 3: The affine pieces of $R(x) = a_i x + b_i$ for $i = 32, \dots, 63$ (left pair) and $i = 128, \dots, 159$ (right pair).

i	$[t_i, t_{i+1}]$	a_i	b_i	i	$[t_i, t_{i+1}]$	a_i	b_i
32	40/7, 220/37	5/2	-10	128	100/7, 330/23	653/40	-321/2
33	220/37, 6	87/20	-21	129	330/23, 620/43	653/40	-321/2
34	6, 260/43	7/2	-27	130	620/43, 29/2	653/40	-321/2
35	260/43, 140/23	7/2	-27	131	29/2, 540/37	693/40	-175
36	140/23, 300/49	7/2	-27	132	540/37, 720/49	767/40	-202
37	300/49, 240/37	91/40	-39/2	133	720/49, 340/23	359/20	-184
38	240/37, 13/2	313/40	-111/2	134	340/23, 640/43	359/20	-184
39	13/2, 280/43	353/40	-62	135	640/43, 15	273/20	-120
40	280/43, 150/23	181/40	-34	136	15, 740/49	64/5	-135
41	150/23, 320/49	181/40	-34	137	740/49, 560/37	463/40	-233/2
42	320/49, 20/3	33/10	-26	138	560/37, 350/23	137/8	-401/2
43	20/3, 340/49	219/40	-81/2	139	350/23, 660/43	137/8	-401/2
44	340/49, 160/23	17/4	-32	140	660/43, 31/2	137/8	-401/2
45	160/23, 300/43	17/4	-32	141	31/2, 760/49	145/8	-216
46	300/43, 7	17/4	-32	142	760/49, 140/9	169/10	-197
47	7, 260/37	17/5	-39	143	140/9, 360/23	667/40	-387/2
48	260/37, 360/49	21/4	-52	144	360/23, 580/37	667/40	-387/2
49	360/49, 170/23	161/40	-43	145	580/37, 680/43	741/40	-445/2
50	170/23, 320/43	161/40	-43	146	680/43, 780/49	569/40	-309/2
51	320/43, 15/2	-11/40	-11	147	780/49, 16	13	-135
52	15/2, 280/37	29/40	-37/2	148	16, 370/23	243/20	-151
53	280/37, 380/49	251/40	-121/2	149	370/23, 600/37	243/20	-151
54	380/49, 180/23	101/20	-51	150	600/37, 700/43	177/10	-241
55	180/23, 340/43	101/20	-51	151	700/43, 800/49	177/10	-241
56	340/43, 8	101/20	-51	152	800/49, 33/2	659/40	-221
57	8, 300/37	21/5	-59	153	33/2, 380/23	699/40	-475/2
58	300/37, 400/49	121/20	-74	154	380/23, 50/3	699/40	-475/2
59	400/49, 190/23	193/40	-64	155	50/3, 820/49	699/40	-475/2
60	190/23, 25/3	193/40	-64	156	820/49, 720/43	65/4	-217
61	25/3, 360/43	193/40	-64	157	720/43, 620/37	239/20	-145
62	360/43, 17/2	21/40	-28	158	620/37, 390/23	69/5	-176
63	17/2, 60/7	61/40	-73/2	159	390/23, 17	69/5	-176

Table 4: The affine pieces of $R(x) = a_i x + b_i$ for $i = 64, \dots, 95$ (left pair) and $i = 160, \dots, 191$ (right pair).

i	$[t_i, t_{i+1}]$	a_i	b_i	i	$[t_i, t_{i+1}]$	a_i	b_i
64	60/7, 320/37	3/10	-26	160	17, 120/7	259/20	-193
65	320/37, 200/23	117/20	-74	161	120/7, 740/43	469/40	-172
66	200/23, 380/43	117/20	-74	162	740/43, 640/37	469/40	-172
67	380/43, 80/9	117/20	-74	163	640/37, 400/23	691/40	-268
68	80/9, 440/49	45/8	-72	164	400/23, 35/2	691/40	-268
69	440/49, 9	22/5	-61	165	35/2, 860/49	731/40	-571/2
70	9, 210/23	71/20	-70	166	860/49, 760/43	341/20	-264
71	210/23, 340/37	71/20	-70	167	760/43, 160/9	51/4	-188
72	340/37, 400/43	27/5	-87	168	160/9, 410/23	501/40	-184
73	400/43, 460/49	11/10	-47	169	410/23, 660/37	501/40	-184
74	460/49, 19/2	-1/8	-71/2	170	660/37, 880/49	115/8	-217
75	19/2, 220/23	7/8	-45	171	880/49, 18	263/20	-195
76	220/23, 360/37	7/8	-45	172	18, 780/43	123/10	-213
77	360/37, 420/43	257/40	-99	173	780/43, 420/23	123/10	-213
78	420/43, 480/49	257/40	-99	174	420/23, 55/3	123/10	-213
79	480/49, 10	26/5	-87	175	55/3, 900/49	123/10	-213
80	10, 500/49	87/20	-97	176	900/49, 680/37	443/40	-381/2
81	500/49, 440/43	25/8	-169/2	177	680/37, 37/2	133/8	-585/2
82	440/43, 380/37	-47/40	-81/2	178	37/2, 800/43	141/8	-311
83	380/37, 240/23	27/40	-119/2	179	800/43, 430/23	533/40	-231
84	240/23, 21/2	27/40	-119/2	180	430/23, 920/49	533/40	-231
85	21/2, 520/49	67/40	-70	181	920/49, 700/37	121/10	-208
86	520/49, 460/43	9/20	-57	182	700/37, 19	279/20	-243
87	460/43, 400/37	9/20	-57	183	19, 820/43	131/10	-262
88	400/37, 250/23	6	-117	184	820/43, 440/23	131/10	-262
89	250/23, 11	6	-117	185	440/23, 940/49	131/10	-262
90	11, 540/49	103/20	-128	186	940/49, 720/37	95/8	-477/2
91	540/49, 100/9	157/40	-229/2	187	720/37, 39/2	697/40	-693/2
92	100/9, 480/43	37/10	-112	188	39/2, 840/43	737/40	-366
93	480/43, 260/23	-3/5	-64	189	840/43, 450/23	113/8	-282
94	260/23, 420/37	-3/5	-64	190	450/23, 960/49	113/8	-282
95	420/37, 80/7	5/4	-85	191	960/49, 20	129/10	-258

Table 5: Affine pieces of the outer integrand $T(y) = b + cy$.

y -interval	b	c
$[1/3, 37/100]$	$-75/8$	8
$[37/100, 37/80]$	$-99/8$	16
$[37/80, 29/60]$	$-569/40$	20
$[29/60, 1/2]$	$-569/40$	20
$[1/2, 37/60]$	$-107/10$	9
$[37/60, 37/40]$	$-251/20$	12
$[37/40, 77/80]$	$-72/5$	14
$[77/80, 1]$	1	-2
$[1, 37/20]$	$-239/20$	-2

B Rational approximation and coefficient height

We prove Corollary 1.3 by controlling the determinant near $\zeta(7)$. The additional input is a relative matrix norm estimate, combined with the stronger decay of Theorem 1.2 for the larger cutoff $M = 100000$. We then record a coefficient height bound.

B.1 The relative matrix norm

A polynomial that is small and nonzero at $\zeta(7)$ might vanish at a nearby rational number. To exclude this, we use the affine dependence of $G_K(X)$ on X and measure its change relative to the positive matrix $G_K(\zeta(7))$.

Put $\xi = \zeta(7)$, $G = G_K(\xi)$, and $M_K = [X]G_K(X)$. Since G is positive definite, the real symmetric matrix

$$B_K = G^{-1/2} M_K G^{-1/2}$$

is defined, with operator norm $\|B_K\|$ equal to the spectral radius.

Proposition B.1. *For all sufficiently large $K \in 40\mathbb{Z}_{>0}$,*

$$\|B_K\| \leq e^{\rho K}, \quad \rho = \frac{14301}{1000}. \quad (123)$$

Proof. Let q be a real polynomial of degree less than h , and write $p(t) = q(K^2 t)$. Fix $0 < a < b$. Keeping only the term $\ell = 1$ in (16), restricting the integral to $y = K\sqrt{t}$ with $a \leq t \leq b$, and bounding every factor separately gives

$$q^T G q \geq \mathcal{A}_K c_a e^{-\mathcal{F}(a,b)K} \int_a^b p(t)^2 dt, \quad (124)$$

where $\mathcal{A}_K = K^{16N-2K+8}$ and

$$c_a = \frac{(2\pi)^6 a^3}{720}, \quad \mathcal{F}(a, b) = 2\pi\sqrt{b} + \log(1+b) - 8\alpha \log a.$$

Indeed, the scaled numerator $D_N(K^2 t)^8$ is at least a^{8N} (since $D_N(u) \geq u^N$ for $u \geq 0$), the denominator is at most $(1+b)^{2K}$ (since $D_K(K^2 t) \leq K^{2K}(1+t)^K$ for $t \leq b$ and $b \leq 1$), and the remaining density is at least $c_a e^{-2\pi K\sqrt{b}}$.

Set $u_j = j/K$. The residue expression for M_K gives

$$q^T M_K q = \mathcal{A}_K \sum_{j=N+1}^K r_j p(-u_j^2)^2, \quad r_j = \frac{u_j^6 \prod_{\ell=1}^N (u_\ell^2 - u_j^2)^7}{\prod_{N < k \leq K} (u_k^2 - u_j^2)}. \quad (125)$$

The numerator has absolute value at most one. Since $u_k + u_j \geq 2\alpha$ for $k \neq j$,

$$|r_j| \leq \frac{(K/(2\alpha))^{h-1}}{(j-N-1)!(K-j)!} \leq \frac{(K/\alpha)^{h-1}}{(h-1)!} \leq \exp\{\lambda K(1 - \log(\alpha\lambda))\}. \quad (126)$$

The Legendre expansion of p on $[a, b]$ gives

$$\max_{-1 \leq t \leq 0} |p(t)|^2 \leq \frac{h^2 \mathcal{R}^{2h}}{b-a} \int_a^b p(t)^2 dt, \quad Y = \frac{2+a+b}{b-a}, \quad \mathcal{R} = Y + \sqrt{Y^2 - 1}. \quad (127)$$

For completeness, the transformed argument has absolute value at most Y , and the Laplace integral for P_j gives $|P_j(s)| \leq \mathcal{R}^j$ there (DLMF 18.10.5). Cauchy–Schwarz in the orthogonal expansion, followed by $\sum_{j=0}^{h-1} (2j+1) = h^2$, proves (127).

Combining (124)–(127) yields

$$\|B_K\| \leq \frac{h^3}{c_a(b-a)} e^{\kappa(a,b)K}, \quad \kappa(a,b) = \mathcal{F}(a,b) + \lambda(1 - \log(\alpha\lambda)) + 2\lambda \log \mathcal{R}.$$

Take $a = 1/18$ and $b = 1/3$. Then $Y = 43/5$ and

$$\kappa = \frac{2\pi}{\sqrt{3}} + \log \frac{4}{3} + \frac{3}{5} \log 18 + \frac{37}{40} \left(1 + \log \frac{1600}{111}\right) + \frac{37}{20} \log \frac{43 + 4\sqrt{114}}{5}.$$

Numerically, $\kappa = 14.2994124425497 \dots < 14301/1000 = 14.301$. The strict inequality is established in Appendix A using the elementary enclosures of §A.2. Its positive margin absorbs the polynomial factor $h^3/(c_a(b-a))$ in the bound above, proving (123). $\square$

B.2 Proof of the approximation bound

Proof of Corollary 1.3. Choose K proportional to $\log b$. The relative norm bound will preserve positivity, while the quadratic decay will outweigh the factor b^h . Use $M = 100000$ and write $Q_K = Q_{K,100000}$. By Theorem 1.2, for all sufficiently large allowed K ,

$$0 < Q_K(\xi) \leq e^{-\epsilon K^2}, \quad \epsilon = 12. \quad (128)$$

This follows from $A_{100000} + \bar{U} < -12$ (see (119)).

Let $c = 75/4$ and $N = 270$. For a positive integer b , choose

$$K = 40 \left\lfloor \frac{c \log b}{40} \right\rfloor, \quad K \leq c \log b < K + 40.$$

Then

$$\rho c = \frac{14301}{1000} \cdot \frac{75}{4} = \frac{1072575}{4000} = 268.14375.$$

Since $N = 270 > \rho c$, we have $\rho c - N = -1.85625 < 0$.

Suppose $a \in \mathbb{Z}$ and $|a/b - \xi| \leq b^{-270}$. Set $\delta = a/b - \xi$. By Proposition B.1 and $K \leq c \log b$,

$$|\delta| \|B_K\| \leq b^{-270} \cdot e^{\rho K} \leq b^{-270+\rho c} = b^{-1.85625} = e^{-1.85625 \log b}.$$

Since $K \leq c \log b$ implies $\log b \geq K/c$, we have

$$|\delta| \|B_K\| \leq e^{-1.85625K/c} = e^{-1.85625 \cdot 4K/75} = e^{-0.099K}.$$

This is $o(K^{-1})$ as $K \rightarrow \infty$, so all eigenvalues of $I + \delta B_K$ are positive for sufficiently large b . Also $\det(I + \delta B_K) \leq (1 + |\delta| \|B_K\|)^h \leq \exp(h|\delta| \|B_K\|) \rightarrow 1$, so the determinant is at most two.

The affine dependence of G_K on X gives

$$Q_K(a/b) = Q_K(\xi) \det(I + \delta B_K).$$

Since $Q_K \in \mathbb{Z}[X]$ has degree h , the number $b^h Q_K(a/b)$ is a nonzero integer, hence at least one in absolute value. Combining with the bound above and using $\log b < (K + 40)/c$,

$$\begin{aligned} \lambda K \log b - \epsilon K^2 &< \lambda K \cdot \frac{K + 40}{c} - 12K^2 = \left(\frac{\lambda}{c} - 12 \right) K^2 + \frac{40\lambda}{c} K \\ &= \left(\frac{37}{750} - 12 \right) K^2 + \frac{37}{18.75} K = -\frac{8963}{750} K^2 + \frac{148}{75} K. \end{aligned}$$

For all sufficiently large K , the right-hand side is at most $-11K^2$. Hence

$$0 < b^h Q_K(a/b) \leq 2e^{-11K^2} < 1$$

for all sufficiently large b , contradicting integrality. Therefore $|\xi - a/b| > b^{-270}$. $\square$

B.3 Coefficient height and primitive normalization

The relative matrix estimate also controls the coefficients, without any further local argument. Let $H(Q)$ be the largest absolute value of a coefficient of Q . If $\beta_1, \dots, \beta_h$ are the eigenvalues of B_K , then

$$Q_K(X) = Q_K(\xi) \prod_{j=1}^h (1 + (X - \xi)\beta_j).$$

The elementary bound $\xi < 2$ and Proposition B.1 imply

$$\log H(Q_K) \leq (\lambda\rho - \epsilon)K^2 + o(K^2) < 2K^2 \quad (129)$$

for all sufficiently large allowed K . Indeed, $\lambda\rho - \epsilon = \frac{37}{40} \cdot \frac{14301}{1000} - 12 = 13.228 - 12 = 1.228 < 2$.

The cutoff enters the explicit normalizing factor, but it can be removed by passing to the primitive polynomial. For $0 \neq A \in \mathbb{Q}[X]$, let $\text{cont}_{\mathbb{Q}}(A) > 0$ be the rational number such that $A/\text{cont}_{\mathbb{Q}}(A)$ has integral coefficients with greatest common divisor one. Put

$$P_K = \Delta_K / \text{cont}_{\mathbb{Q}}(\Delta_K).$$

For every admissible K, M , there is a positive integer $a_{K,M}$ with $Q_{K,M} = a_{K,M} P_K$. Indeed, a rational multiple of a primitive integer polynomial has integer coefficients only when the multiplier is a positive integer. Thus $P_K(\xi) > 0$, and the upper bounds proved above for $Q_{K,M}(\xi)$ hold for $P_K(\xi)$ as well.

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